CHAPTER 18

MORE SPECIAL FUNCTIONS

In this chapter we shall study four sets of orthogonal polynomials: Hermite, Laguerre, and Chebyshev[1] of the first and second kinds. Although these four sets are of less importance in mathematical physics than are the Bessel and Legendre functions of Chapters 14 and 15, they are used and therefore deserve attention. For example, Hermite polynomials occur in solutions of the simple harmonic oscillator of quantum mechanics and Laguerre polynomials in wave functions of the hydrogen atom. Because the general mathematical techniques duplicate those used for Bessel and Legendre functions, the development of these functions is only outlined. Detailed proofs are for the most part left to the reader.

The sets of polynomials treated in this chapter can be related to the more general quantities known as **hypergeometric** and **confluent hypergeometric** functions (solutions of the hypergeometric ODE). For practical reasons we defer most discussion of these relationships until we have had an opportunity to define the hypergeometric functions and the associated nomenclature. The benefit accruing from the connection to hypergeometric functions is that the hypergeometic recurrence formulas and other general properties translate into useful relationships for the polynomial sets that we are presently studying.

We conclude the chapter with a short section on elliptic integrals. Although the importance of this subject has declined as the power of computers has increased, there are some physical problems for which they are useful and it is not yet time to eliminate them from this text.

18.1 HERMITE FUNCTIONS

We start by identifying **Hermite functions** as solutions of the **Hermite** ODE,

$$H_n''(x) - 2x H_n'(x) + 2n H_n(x) = 0. \tag{18.1}$$

[1]This is the spelling choice of AMS-55 (for the complete reference, see Abramowitz in Additional Readings). However, various names, such as Tschebyscheff, are encountered in the literature.

Mathematical Methods for Physicists.

Here n is a parameter. When $n \geq 0$ is integral, this ODE will have a solution $H_n(x)$ which is a polynomial of degree n; these solutions are known as **Hermite polynomials**.

In the presence of appropriate boundary conditions, the Hermite ODE is a Sturm-Liouville system; polynomial solutions to such ODEs was the topic of Section 12.1. We showed there, in Example 12.1.1, that the Hermite polynomials could be generated from their **Rodrigues formula**, Eq. (12.17), and that, in turn, a Rodrigues formula can be obtained from the underlying ODE. We also showed in that same section how we can go from the Rodrigues formula to a generating function for a given polynomial set, presenting in Table 12.1 a list of generating functions that could be found in this way. That list included the following generating function for the Hermite polynomials:

$$g(x,t) = e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}. \tag{18.2}$$

Here we elect not to depend on the analysis of Section 12.1 but rather to take the viewpoint that Eq. (18.2) can be regarded as a **definition** of the Hermite polynomials, thereby making the present analysis completely self-contained. Accordingly, we proceed by verifying that these polynomials satisfy the Hermite ODE, have the expected Rodrigues formula, and exhibit the other properties that can be developed starting from the generating function.

Recurrence Relations

Note the absence of a superscript, which distinguishes Hermite polynomials from the unrelated Hankel functions. From the generating function we find that the Hermite polynomials satisfy the recurrence relations

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \tag{18.3}$$

and

$$H_n'(x) = 2nH_{n-1}(x). \tag{18.4}$$

The Hermite polynomials were used in Example 12.1.2 as a detailed illustration of the method for obtaining recurrence formulas from generating functions; we summarize the process here. By differentiating the generating function formula with respect to t we obtain

$$\frac{\partial g}{\partial t} = (-2t+2x)e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_{n+1}(x)\frac{t^n}{n!}, \quad \text{or}$$

$$-2\sum_{n=0}^{\infty} H_n(x)\frac{t^{n+1}}{n!} + 2x\sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!} = \sum_{n=0}^{\infty} H_{n+1}(x)\frac{t^n}{n!}.$$

Because this equation must be satisfied separately for each power of t, we arrive at Eq. (18.3). Similarly, differentiation with respect to x leads to

$$\frac{\partial g}{\partial x} = 2te^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n'(x)\frac{t^n}{n!} = 2\sum_{n=0}^{\infty} H_n(x)\frac{t^{n+1}}{n!},$$

from which we can obtain Eq. (18.4).

The Maclaurin expansion of the generating function

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{(2tx - t^2)^n}{n!} = 1 + (2tx - t^2) + \cdots \tag{18.5}$$

gives $H_0(x) = 1$ and $H_1(x) = 2x$, and then the recursion formula, Eq. (18.3), permits the construction of any $H_n(x)$ desired. For convenient reference the first several Hermite polynomials are listed in Table 18.1 and presented graphically in Fig. 18.1.

Special Values

Special values of the Hermite polynomials follow from the generating function for $x = 0$:

$$e^{-t^2} = \sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!} = \sum_{n=0}^{\infty} H_n(0) \frac{t^n}{n!},$$

Table 18.1 Hermite Polynomials

$H_0(x) = 1$
$H_1(x) = 2x$
$H_2(x) = 4x^2 - 2$
$H_3(x) = 8x^3 - 12x$
$H_4(x) = 16x^4 - 48x^2 + 12$
$H_5(x) = 32x^5 - 160x^3 + 120x$
$H_6(x) = 64x^6 - 480x^4 + 720x^2 - 120$

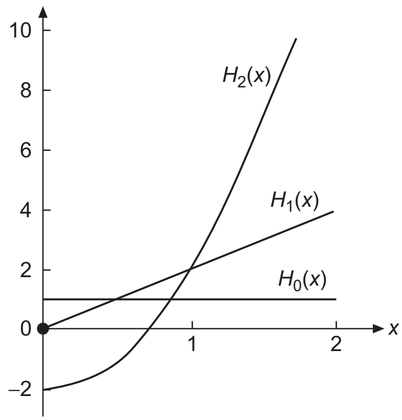

FIGURE 18.1 Hermite polynomials.

that is,

$$H_{2n}(0) = (-1)^n \frac{(2n)!}{n!}, \quad H_{2n+1}(0) = 0, \quad n = 0, 1, \cdots. \tag{18.6}$$

We also obtain from the generating function the important parity relation

$$H_n(x) = (-1)^n H_n(-x) \tag{18.7}$$

by noting that Eq. (18.3) yields

$$g(-x, -t) = \sum_{n=0}^{\infty} H_n(-x) \frac{(-t)^n}{n!} = g(x, t) = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

Hermite ODE

If we substitute the recursion formula Eq. (18.4) into Eq. (18.3), we can eliminate the index $n-1$, obtaining

$$H_{n+1}(x) = 2x H_n(x) - H_n'(x).$$

If we differentiate this recurrence relation and substitute Eq. (18.4) for the index $n+1$, we find

$$H_{n+1}'(x) = 2(n+1) H_n(x) = 2H_n(x) + 2x H_n'(x) - H_n''(x),$$

which can be rearranged to the second-order Hermite ODE, Eq. (18.1). This completes the process of establishing the identification of the Hermite polynomials obtained from the generating function as solutions of the Hermite ODE.

Rodrigues Formula

A simple way to generate the Rodrigues formula for the Hermite polynomials starts from the observations that

$$g(x, t) = e^{-t^2 + 2tx} = e^{x^2} e^{-(t-x)^2} \quad \text{and} \quad \frac{\partial}{\partial t} e^{-(t-x)^2} = -\frac{\partial}{\partial x} e^{-(t-x)^2}.$$

We note that n-fold differentiation of the generating function formula, Eq. (18.2), followed by setting $t = 0$, yields

$$\left. \frac{\partial^n}{\partial t^n} g(x, t) \right|_{t=0} = H_n(x),$$

and we can therefore obtain the Rodrigues formula as

$$H_n(x) = \left. \frac{\partial^n}{\partial t^n} g(x, t) \right|_{t=0} = e^{x^2} \left. \frac{\partial^n}{\partial t^n} e^{-(t-x)^2} \right|_{t=0} = (-1)^n e^{x^2} \left. \frac{\partial^n}{\partial x^n} e^{-(t-x)^2} \right|_{t=0}$$

$$= (-1)^n e^{x^2} \frac{\partial^n}{\partial x^n} e^{-x^2}. \tag{18.8}$$

Series Expansion

Starting from the Maclaurin expansion, Eq. (18.5), we can derive our Hermite polynomial $H_n(x)$ in series form: Using the binomial expansion of $(2x-t)^\nu$, we initially get

$$e^{-t^2+2tx} = \sum_{\nu=0}^{\infty} \frac{t^\nu}{\nu!}(2x-t)^\nu = \sum_{\nu=0}^{\infty} \frac{t^\nu}{\nu!} \sum_{s=0}^{\nu} \binom{\nu}{s} (2x)^{\nu-s}(-t)^s$$

$$= \sum_{\nu=0}^{\infty} \sum_{s=0}^{\nu} \frac{t^{\nu+s}}{(\nu+s)!} \frac{(-1)^s(\nu+s)!\,(2x)^{\nu-s}}{(\nu-s)!\,s!}.$$

Changing the first summation index from ν to $n=\nu+s$, and noting that this change causes the s summation to range from zero to $[n/2]$, the largest integer less than or equal to $n/2$, our expansion takes the form

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{t^n}{n!} \sum_{s=0}^{[n/2]} \frac{(-1)^s n!}{(n-2s)!\,s!}(2x)^{n-2s},$$

from which we can read out the formula for H_n:

$$H_n(x) = \sum_{s=0}^{[n/2]} \frac{(-1)^s n!}{(n-2s)!\,s!}(2x)^{n-2s}. \tag{18.9}$$

Finally, we note that $H_n(x)$ can be written as a Schlaefli integral. Comparing with Eq. (12.18),

$$H_n(x) = \frac{n!}{2\pi i} \oint t^{-n-1} e^{-t^2+2tx}\, dt. \tag{18.10}$$

Orthogonality and Normalization

The orthogonality of the Hermite polynomials is demonstrated by identifying them as arising in a Sturm-Liouville system. The Hermite ODE, however, is clearly **not** self-adjoint, but can be made so by multiplying it by $\exp(-x^2)$ (see Exercise 8.2.2). With $\exp(-x^2)$ as a weighting factor, we obtain the orthogonality integral

$$\int_{-\infty}^{\infty} H_m(x)H_n(x)\,e^{-x^2}dx = 0, \quad m \neq n. \tag{18.11}$$

The interval $(-\infty, \infty)$ is chosen to obtain the Hermitian operator boundary conditions (see Section 8.2).

It is sometimes convenient to absorb the weighting function into the Hermite polynomials. We may define

$$\varphi_n(x) = e^{-x^2/2} H_n(x), \tag{18.12}$$

with $\varphi_n(x)$ no longer a polynomial. Substitution into Eq. (18.1) yields the differential equation for $\varphi_n(x)$,

$$\varphi_n''(x) + (2n + 1 - x^2)\varphi_n(x) = 0. \tag{18.13}$$

Equation (18.13) is self-adjoint, and the solutions $\varphi_n(x)$ are orthogonal on the interval $-\infty < x < \infty$ with a unit weighting function.

We still need to normalize these functions. One approach is to combine two instances of the generating function formula (using variables s and t), after which we multiply by e^{-x^2} and integrate over x from $-\infty$ to ∞. These steps yield

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-s^2+2sx} e^{-t^2+2tx} dx = \sum_{m,n=0}^{\infty} \frac{s^m t^n}{m!\,n!} \int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx. \tag{18.14}$$

We next note that the exponentials on the left-hand side of Eq. (18.14) can be combined into $e^{2st} e^{-(x-s-t)^2}$, after which the integral can be evaluated:

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-s^2+2sx} e^{-t^2+2tx} dx = e^{2st} \int_{-\infty}^{\infty} e^{-(x-s-t)^2} dx = \pi^{1/2} e^{2st}.$$

Inserting this result into Eq. (18.14) after expanding it in a power series, we get

$$\pi^{1/2} e^{2st} = \pi^{1/2} \sum_{n=0}^{\infty} \frac{2^n s^n t^n}{n!} = \sum_{m,n=0}^{\infty} \frac{s^m t^n}{m!\,n!} \int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx.$$

By equating coefficients of equal powers of s and t, we both confirm the orthogonality and obtain the normalization integral

$$\int_{-\infty}^{\infty} e^{-x^2} \Big[H_n(x) \Big]^2 dx = 2^n \pi^{1/2} n!. \tag{18.15}$$

Exercises

18.1.1 Assume that the Hermite polynomials are known to be solutions of the Hermite ODE, Eq. (18.1). Assume further that the recurrence relation, Eq. (18.3), and the values of $H_n(0)$ are also known. Given the existence of a generating function

$$g(x, t) = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!},$$

(a) Differentiate $g(x, t)$ with respect to x and using the recurrence relation develop a first-order PDE for $g(x, t)$.

(b) Integrate with respect to x, holding t fixed.
(c) Evaluate $g(0,t)$ using the known values of $H_n(0)$.
(d) Finally, show that $g(x,t) = \exp(-t^2 + 2tx)$.

18.1.2 In developing the properties of the Hermite polynomials, start at a number of different points, such as:

1. Hermite's ODE, Eq. (18.1),
2. Rodrigues's formula, Eq. (18.8),
3. Integral representation, Eq. (18.10),
4. Generating function, Eq. (18.2),
5. Gram-Schmidt construction of a complete set of orthogonal polynomials over $(-\infty, \infty)$ with a weighting factor of $\exp(-x^2)$ (Section 5.2).

Outline how you can go from any one of these starting points to all the other points.

18.1.3 Prove that $|H_n(x)| \le |H_n(ix)|$.

18.1.4 Rewrite the series form of $H_n(x)$, Eq. (18.9), as an **ascending** power series.

$$\text{ANS.} \quad H_{2n}(x) = (-1)^n \sum_{s=0}^{n} (-1)^{2s} (2x)^{2s} \frac{(2n)!}{(2s)!(n-s)!},$$

$$H_{2n+1}(x) = (-1)^n \sum_{s=0}^{n} (-1)^s (2x)^{2s+1} \frac{(2n+1)!}{(2s+1)!(n-s)!}.$$

18.1.5 (a) Expand x^{2r} in a series of even-order Hermite polynomials.
(b) Expand x^{2r+1} in a series of odd-order Hermite polynomials.

$$\text{ANS.} \quad \text{(a) } x^{2r} = \frac{(2r)!}{2^{2r}} \sum_{n=0}^{r} \frac{H_{2n}(x)}{(2n)!(r-n)!}$$

$$\text{(b) } x^{2r+1} = \frac{(2r+1)!}{2^{2r+1}} \sum_{n=0}^{r} \frac{H_{2n+1}(x)}{(2n+1)!(r-n)!}, \qquad r = 0, 1, 2, \ldots.$$

Hint. Use a Rodrigues representation and integrate by parts.

18.1.6 Show that

$$\text{(a)} \quad \int_{-\infty}^{\infty} H_n(x) \exp\left[-\frac{x^2}{2}\right] dx = \begin{cases} 2\pi n!/(n/2)!, & n \text{ even} \\ 0, & n \text{ odd.} \end{cases}$$

$$\text{(b)} \quad \int_{-\infty}^{\infty} x H_n(x) \exp\left[-\frac{x^2}{2}\right] dx = \begin{cases} 0, & n \text{ even} \\ 2\pi \dfrac{(n+1)!}{((n+1)/2)!}, & n \text{ odd.} \end{cases}$$

18.1.7 (a) Using the Cauchy integral formula, develop an integral representation of $H_n(x)$ based on Eq. (18.2) with the contour enclosing the point $z = -x$.

$$ANS. \quad H_n(x) = \frac{n!}{2\pi i} e^{x^2} \oint \frac{e^{-z^2}}{(z+x)^{n+1}}\, dz.$$

(b) Show by direct substitution that this result satisfies the Hermite equation.

18.2 Applications of Hermite Functions

One of the most important applications of Hermite functions in physics arises from the fact that the functions $\varphi_n(x)$ of Eq. (18.12) are the eigenstates of the quantum-mechanical simple harmonic oscillator, which describes motion subject to a quadratic (also known as a **harmonic** or a **Hooke's-law**) potential. This fact causes Hermite polynomials not only to appear in elementary quantum-mechanics problems, but also in analyses of the vibrational states of molecules, where the lowest-order description of the interatomic potential is harmonic. In view of the importance of these topics, we now proceed to examine them in some detail.

Simple Harmonic Oscillator

The quantum mechanical simple harmonic oscillator is governed by a Schrödinger equation of the form

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(z)}{dz^2} + \frac{k}{2} z^2 \psi(z) = E\psi(z), \tag{18.16}$$

where m is the mass of the oscillator, k is the force constant for its Hooke's law force directed toward $z = 0$, $\hbar$ is Planck's constant divided by 2π, and E is an eigenvalue giving the energy of the oscillator. Equation (18.16) is to be solved subject to the boundary condition that $\psi(z)$ vanish at $z = \pm\infty$. It is convenient to make a change of variable that eliminates the various constants from the equation, and we therefore make the substitutions

$$z = \frac{\hbar^{1/2} x}{(km)^{1/4}}, \qquad \frac{k}{2} z^2 = \frac{\hbar}{2}\sqrt{\frac{k}{m}}\, x^2, \qquad \frac{\hbar^2}{2m}\frac{d^2}{dz^2} = \frac{\hbar}{2}\sqrt{\frac{k}{m}}\frac{d^2}{dx^2},$$

which converts Eq. (18.16) into

$$-\frac{1}{2}\frac{d^2\varphi(x)}{dx^2} + \frac{x^2}{2}\varphi(x) = \lambda\varphi(x), \tag{18.17}$$

with boundary conditions at $x = \pm\infty$. The eigenvalue λ in this equation is related to E by

$$E = \hbar\lambda\sqrt{\frac{k}{m}}. \tag{18.18}$$

The solutions of Eq. (18.17) that satisfy the boundary conditions can now be identified as given by Eq. (18.13), and we can identify λ_n, the eigenvalue of Eq. (18.17) corresponding to $\varphi_n(x)$, as having the value $n + \frac{1}{2}$. Turning to Eq. (18.12), and expressing x in terms of the original variable z, the eigenstates of Eq. (18.16) can be characterized (including a normalization constant N_n) as

$$\psi_n(z) = N_n e^{-(\alpha z)^2/2} H_n(\alpha z), \quad E_n = (n + \tfrac{1}{2})\hbar\sqrt{\frac{k}{m}}, \quad \alpha = \frac{\hbar^{1/2}}{(km)^{1/4}}, \tag{18.19}$$

with n restricted to the integer values 0, 1, 2, $\cdots$. The normalization constant can be deduced from Eq. (18.15). Noting that the normalization integral is to be over the variable z, we find it to be

$$N_n = \left(\frac{\alpha}{2^n \pi^{1/2} n!}\right)^{1/2}. \tag{18.20}$$

It is of interest to examine a few of the eigenstates of this oscillator problem. For reference, a classical oscillator of mass m and force constant k will have the angular oscillation frequency

$$\omega_{\text{class}} = \sqrt{\frac{k}{m}},$$

and can have an arbitrary energy of oscillation, while our quantum oscillator is restricted to oscillation energies $(n + \frac{1}{2})\hbar\omega_{\text{class}}$, with n a nonnegative integer. We note that the quantum oscillator must have at least the total energy $\frac{1}{2}\hbar\omega_{\text{class}}$; this is usually referred to as its **zero-point energy** and is a consequence of the fact that its spatial distribution must be described by a wave function of finite extent.

The three lowest-energy eigenfunctions of the quantum oscillator are shown in Fig. 18.2. We note that these wave functions predict a position distribution that extends to $\pm\infty$, albeit with exponentially decaying amplitude for larger $|z|$. The corresponding classical oscillator will have excursions in z that are strictly bounded by $kz_{\max}^2/2 = E$, where E can be assigned any value greater than or equal to zero. We have marked in Fig. 18.2 the excursion range of a classical oscillator with an energy equal to the eigenvalue of the quantum oscillator; note that the exponential decay of the quantum wave function begins at the ends of the classical range.

Operator Approach

While the analysis of the preceding subsection is straightforward and provides a complete set of eigenstates for the simple quantum oscillator, additional insight can be obtained by an alternative approach that uses the commutation and other algebraic properties of the quantum-mechanical operators. Our starting point for this development is the recognition that the differential operator $-d^2/dx^2$ of Eq. (18.17) arose as a representation of the dynamical quantity p^2, where (in units with $\hbar = 1$) $p \longleftrightarrow -i\, d/dx$. Then our Schrödinger equation of Eq. (18.17) can be written

$$\mathcal{H}\varphi = \frac{p^2 + x^2}{2}\varphi = \lambda\varphi, \tag{18.21}$$

where $\mathcal{H}$ is the Hamiltonian operator, with eigenvalues λ.

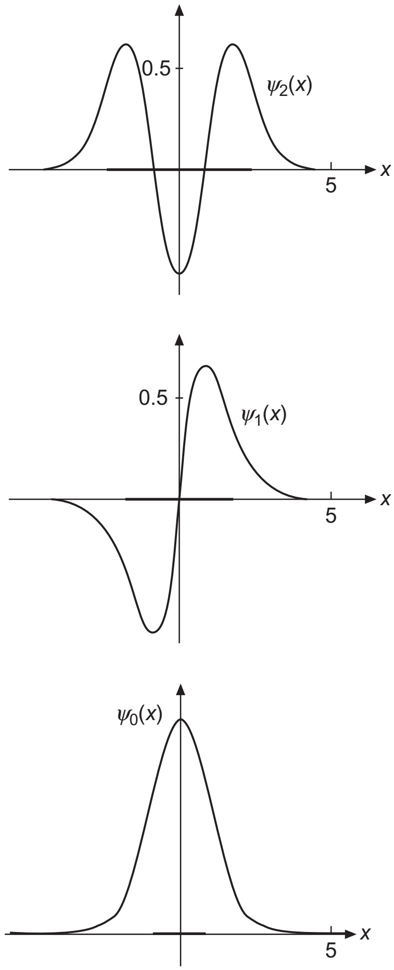

FIGURE 18.2 Quantum mechanical oscillator wave functions. The heavy bar on the x-axis indicates the allowed range of the classical oscillator with the same total energy.

The key to an approach starting from Eq. (18.21) is that x and p satisfy the basic commutation relation

$$[x, p] = xp - px = i, \tag{18.22}$$

a result discussed in detail in the analysis leading to Eq. (5.43). In fact, if we proceed under the assumption that Eq. (18.22) is all that we know about x and p, there is the additional advantage that any results we obtain will be more general than those from our original oscillator problem in ordinary space. This observation underlies much recent work in which physical theory has evolved in more abstract directions.

With a knowledge of the way in which angular momentum theory was developed in terms of raising and lowering operators, one can easily motivate a somewhat similar procedure here, by defining the two operators

$$a = \frac{1}{\sqrt{2}}(x + ip), \quad a^\dagger = \frac{1}{\sqrt{2}}(x - ip). \tag{18.23}$$

Since we typically use a to denote a constant, we remind the reader that in the present development it is an operator (involving x and d/dx). With suitable Sturm-Liouville boundary conditions, x and p are both Hermitian. But the presence of the imaginary unit i causes a not to be Hermitian, and (as indicated by the notation) changing the sign of the term ip converts a into its adjoint, $a^\dagger$.

Our first use of Eq. (18.23) is to form $a^\dagger a$ and $aa^\dagger$:

$$a^\dagger a = \frac{1}{2}(x - ip)(x + ip) = \frac{1}{2}(x^2 + p^2) + \frac{i}{2}(xp - px) = \mathcal{H} + \frac{i}{2}[x, p] = \mathcal{H} - \frac{1}{2},$$

$$aa^\dagger = \frac{1}{2}(x + ip)(x - ip) = \frac{1}{2}(x^2 + p^2) - \frac{i}{2}(xp - px) = \mathcal{H} - \frac{i}{2}[x, p] = \mathcal{H} + \frac{1}{2}.$$

From these equations we obtain the useful formulas

$$\mathcal{H} = a^\dagger a + \tfrac{1}{2}, \tag{18.24}$$

$$[a, a^\dagger] = aa^\dagger - a^\dagger a = 1, \tag{18.25}$$

and therefrom

$$[\mathcal{H}, a] = [a^\dagger a + \tfrac{1}{2}, a] = [a^\dagger a, a] = a^\dagger aa - aa^\dagger a = (a^\dagger a - aa^\dagger)a = -a. \tag{18.26}$$

Applying $[\mathcal{H}, a]$ to an eigenfunction φ_n with eigenvalue λ_n (assumed not yet known), we write

$$[\mathcal{H}, a]\varphi_n = H(a\varphi_n) - aH\varphi_n = H(a\varphi_n) - \lambda_n(a\varphi_n) = -(a\varphi_n),$$

which we easily rearrange to the form

$$\mathcal{H}(a\varphi_n) = (\lambda_n - 1)(a\varphi_n). \tag{18.27}$$

Equation (18.27) shows that we can interpret a as a **lowering** operator that converts an eigenfunction with eigenvalue λ_n into another eigenfunction that has eigenvalue $\lambda_n - 1$. A similar analysis, left to the reader, shows that from the commutator $[\mathcal{H}, a^\dagger]$ we find $a^\dagger$ to be a **raising** operator, according to

$$\mathcal{H}(a^\dagger\varphi_n) = (\lambda_n + 1)(a^\dagger\varphi_n). \tag{18.28}$$

These formulas show that, given any eigenfunction φ_n, we can construct a **ladder** of eigenstates whose eigenvalues differ by unit steps. The only limitation that would terminate the construction of an infinite ladder would be the possibility that for some φ_n, either $a\varphi_n$ or $a^\dagger\varphi_n$ might be zero.

To investigate the circumstances under which $a\varphi_n$ might vanish, let's form the scalar product $\langle a\varphi_n|a\varphi_n\rangle$. We find

$$\langle a\varphi_n|a\varphi_n\rangle = \langle\varphi_n|a^\dagger a|\varphi_n\rangle = \langle\varphi_n|\mathcal{H} - \tfrac{1}{2}|\varphi_n\rangle = \langle\varphi_n|\lambda_n - \tfrac{1}{2}|\varphi_n\rangle. \tag{18.29}$$

Equation (18.29) shows that only if $\lambda_n = \frac{1}{2}$ will we have $a\varphi_n = 0$. That equation also shows that if $\lambda_n < \frac{1}{2}$ we have the mathematical inconsistency that the norm of $a\varphi_n$ is predicted to be negative. These observations together imply that the only possible values of λ_n are positive half-integers, as otherwise by repeated application of the lowering operator a we can move to a λ value prohibited by Eq. (18.29). We leave to the reader the verification that the application of the raising operator $a^\dagger$ to any valid φ_n produces a new eigenfunction $a^\dagger\varphi_n$ with a positive norm.

Our overall conclusion is that any system with a Hamiltonian of the form given by Eq. (18.21), whether or not represented by an ODE in ordinary space, will have eigenstates whose eigenvalues form a ladder of unit spacing, with the smallest eigenvalue equal to $\frac{1}{2}$. This makes it natural to label the states φ_n by integers $n \geq 0$, and therefore to write

$$\mathcal{H}\varphi_n = \lambda_n\varphi_n, \quad \lambda_n = n + \tfrac{1}{2}, \quad n = 0,\ 1,\ 2\cdots, \tag{18.30}$$

in agreement with what we found from our original approach; compare Eq. (18.19).

Before leaving this exercise in operator algebra, it may be worth noting that the notion of raising and lowering operators also arises in contexts where the states thereby reached can be interpreted as those containing different numbers of particles (or **quasiparticles**, a physics jargon that refers to objects, such as photons, whose population is easily changed by interaction with their surroundings). In such contexts, a raising operator is then often referred to as a **creation operator**, with a lowering operator then called an **annihilation** (or sometimes a **destruction**) operator. Obviously these terms have to be interpreted with an understanding of the underlying physics.

Returning to the description of p as a differential operator, the equation $a\varphi_0 = 0$ can be identified as a differential equation satisfied by the ground (lowest-energy) state of our oscillator. More specifically,

$$\sqrt{2}\,a\varphi_0 = (x + ip)\varphi_0 = \left[x + i\left(-i\frac{d}{dx}\right)\right]\varphi_0 = \left[x + \frac{d}{dx}\right]\varphi_0 = 0, \tag{18.31}$$

which has the advantage of being a **first-order** ODE. This ODE is separable, and can be integrated:

$$\frac{d\varphi_0}{\varphi_0} = -x\,dx, \quad \ln\varphi_0 = -\frac{x^2}{2} + \ln c_0, \quad \varphi_0 = c_0\,e^{-x^2/2},$$

in agreement with our previous analysis.

Eigenstates for arbitrary n can now be generated by repeated application of $a^\dagger$ to φ_0. Doing so is left as an exercise.

Molecular Vibrations

In the dynamics and spectroscopy of molecules in the Born-Oppenheimer approximation, the motion of a molecule is separated into electronic, vibrational, and rotational motion. In treating the vibrational motion, the departure of nuclei from their equilibrium positions is to lowest order described by a quadratic potential, and the resulting oscillations are identified as **harmonic**. These harmonic motions can be treated as coupled simple harmonic oscillators, and we can decouple the individual nuclear motions by making a transformation to **normal** coordinates, as was illustrated in Example 6.5.2. In this harmonic oscillation limit, the vibrational wave functions have the form given in the preceding subsection, and the computation of properties associated with these wave functions then involve integrals in which products of Hermite functions appear.

The simplest integrals occurring in vibrational problems are of the form

$$\int_{-\infty}^{\infty} x^r e^{-x^2} H_n(x) H_m(x) dx.$$

Examples for $r = 1$ and $r = 2$ (with $n = m$) are included in the exercises at the end of this section. A large number of other examples can be found in the work by Wilson, Decius, and Cross.[2] Some of the vibrational properties of molecules require the evaluation of integrals containing as many as four Hermite functions. In the remainder of this subsection we illustrate some of the possibilities and the associated mathematical procedures.

Example 18.2.1 Threefold Hermite Formula

Consider the following integral involving three Hermite polynomials

$$I_3 \equiv \int_{-\infty}^{\infty} e^{-x^2} H_{m_1}(x) H_{m_2}(x) H_{m_3}(x) dx, \tag{18.32}$$

where $N_i \geq 0$ are integers. The formula (due to E. C. Titchmarsh, *J. Lond. Math. Soc.* **23**: 15 (1948); see Gradshteyn and Ryzhik, p. 804, in Additional Readings) generalizes the I_2 case needed for the orthogonality and normalization of Hermite polynomials. To start, we note that the integrand of I_3 will be even if the index sum $m_1 + m_2 + m_3$ is even, and odd if that index sum is odd, so I_3 will vanish unless $m_1 + m_2 + m_3$ is even. In addition, we see that if the product $H_{m_1} H_{m_2}$ is expanded and written as a sum of Hermite polynomials, the resulting polynomial of largest index will be $H_{m_1+m_2}$, so I_3 will vanish due to orthogonality unless $m_1 + m_2$ is at least as large as m_3. This condition must continue to hold if the roles of the m_i are permuted; a convenient way of summarizing these observations is to state that the m_i must satisfy a **triangle** condition. Both the even index sum and the triangle condition parallel similar conditions on integrals of Legendre polynomials which we encountered in Section 16.3 and discussed in detail at Eq. (16.85).

[2] E. B. Wilson, Jr., J. C. Decius, and P. C. Cross, *Molecular Vibrations*, New York: McGraw-Hill (1955), reprinted, Dover (1980).

To derive I_3, we start with the product of three generating functions of Hermite polynomials, multiply by e^{-x^2}, and integrate over x:

$$Z_3 \equiv \int_{-\infty}^{\infty} e^{-x^2} \prod_{j=1}^{3} e^{2xt_j - t_j^2}\, dx = \int_{-\infty}^{\infty} e^{-(t_1+t_2+t_3-x)^2 + 2(t_1t_2+t_1t_3+t_2t_3)} dx$$

$$= \sqrt{\pi}\, e^{2(t_1t_2+t_1t_3+t_2t_3)} = \sqrt{\pi} \sum_{N=0}^{\infty} \frac{2^N}{N!} \sum_{\substack{n_1,n_2,n_3 \geq 0 \\ n_1+n_2+n_3=N}} \frac{N!}{n_1!\,n_2!\,n_3!}\, t_1^{n_2+n_3} t_2^{n_1+n_3} t_3^{n_1+n_2}. \tag{18.33}$$

In reaching Eq. (18.33), we recognized the x integration as an error integral, Eq. (1.148), and then expanded the resulting exponential, first as a power series in $w = 2(t_1t_2 + t_1t_3 + t_2t_3)$, and then expanding the powers of w by the generalization of the binomial theorem given as Eq. (1.80). Note that the index for the power of $t_i t_j$ in the polynomial expansion was designated n_k, where $i,\ j,\ k$ are (in some order) $1, 2, 3$.

We next expand the generating functions in terms of Hermite polynomials and set the result equal to a slightly simplified version of the expression just obtained for Z_3:

$$Z_3 = \sum_{m_1,m_2,m_3=0}^{\infty} \frac{t_1^{m_1} t_2^{m_2} t_3^{m_3}}{m_1!\,m_2!\,m_3!} \int_{-\infty}^{\infty} e^{-x^2} H_{m_1}(x) H_{m_2}(x) H_{m_3}(x) dx$$

$$= \sqrt{\pi} \sum_{n_1,n_2,n_3=0}^{\infty} \frac{2^N\, t_1^{n_2+n_3} t_2^{n_1+n_3} t_3^{n_1+n_2}}{n_1!\,n_2!\,n_3!}, \tag{18.34}$$

with $N = n_1 + n_2 + n_3$. In Eq. (18.34) we now equate the coefficients of equal powers of the t_j, finding that $m_1 = n_2 + n_3$, $m_2 = n_1 + n_3$, $m_3 = n_1 + n_2$, that

$$N = \frac{m_1 + m_2 + m_3}{2},$$

and that $n_1 = N - m_1$, $n_2 = N - m_2$, $n_3 = N - m_3$. From the coefficients of $t_1^{m_1} t_2^{m_2} t_3^{m_3}$, we obtain the final result

$$I_3 = \frac{\sqrt{\pi}\ 2^N\, m_1!\,m_2!\,m_3!}{(N - m_1)!\,(N - m_2)!\,(N - m_3)!}. \tag{18.35}$$

Equation (18.35) explicitly reflects the necessity of the triangle condition. If it is not satisfied but the sum of the m_i is even, at least one of the factorials in the denominator of Eq. (18.35) will have a negative integer argument, thereby causing I_3 to be zero. The requirement that the sum of the m_i be even is not explicit in the form of Eq. (18.35), but the formula for I_3 is restricted to that case because the right-hand side of Eq. (18.34) only contains terms in which the sum of the powers of the t_i is even. ■

Hermite Product Formula

The integrals I_m with $m > 3$ can be obtained in closed form, but as finite sums. The starting point for that analysis is a formula for the product of two Hermite polynomials due to

E. Feldheim, *J. Lond. Math. Soc.* **13**: 22 (1938). To derive Feldheim's formula, we can start from a product of two generating functions, written as

$$e^{2x(t_1+t_2)-t_1^2-t_2^2} = \sum_{m_1,m_2=0}^{\infty} H_{m_1}(x)H_{m_2}(x)\,\frac{t_1^{m_1}}{m_1!}\,\frac{t_2^{m_2}}{m_2!}$$

$$= e^{2x(t_1+t_2)-(t_1+t_2)^2}\, e^{2t_1t_2} = \sum_{n=0}^{\infty} H_n(x)\frac{(t_1+t_2)^n}{n!}\sum_{\nu=0}^{\infty}\frac{(2t_1t_2)^{\nu}}{\nu!}.$$

Applying the binomial expansion to $(t_1+t_2)^n$ and then comparing like powers of t_1 and t_2 in the two lines of the above equation, we find

$$H_{m_1}(x)H_{m_2}(x) = \sum_{\nu=0}^{\min(m_1,m_2)} H_{m_1+m_2-2\nu}(x)\frac{m_1!\,m_2!\,2^{\nu}}{\nu!\,(m_1+m_2-2\nu)!}\binom{m_1+m_2-2\nu}{m_1-\nu}$$

$$= \sum_{\nu=0}^{\min(m_1,m_2)} H_{m_1+m_2-2\nu}(x)\,2^{\nu}\nu!\binom{m_1}{\nu}\binom{m_2}{\nu}. \tag{18.36}$$

For $\nu = 0$ the coefficient of $H_{N_1+N_2}$ is obviously unity. Special cases, such as

$$H_1^2 = H_2 + 2,\;\; H_1H_2 = H_3 + 4H_1,\;\; H_2^2 = H_4 + 8H_2 + 8,\;\; H_1H_3 = H_4 + 6H_2$$

can be derived from Table 13.1 and agree with the general twofold product formula.

The product formula has been generalized to products of $m > 2$ Hermite polynomials, thereby providing a new way of evaluating the integrals I_m. For details we refer the reader to work by Liang, Weber, Hayashi, and Lin.[3]

Example 18.2.2 FOURFOLD HERMITE FORMULA

An important application of the Hermite product formula is a newly reported evaluation of the integral I_4 containing a product of four Hermite polynomials. The analysis is that of one of the present authors and his colleagues.[3]

The integral we are about to study is of the form

$$I_4 = \int_{-\infty}^{\infty} e^{-x^2} H_{m_1}(x)H_{m_2}(x)H_{m_3}(x)H_{m_4}(x)\,dx. \tag{18.37}$$

It is convenient to order the indices of the Hermite polynomials so that $m_1 \ge m_2 \ge m_3 \ge m_4$. Our approach will be to apply the product formula to $H_{m_1}H_{m_2}$ and to $H_{m_3}H_{m_4}$, thereby

[3]K. K. Liang, H. J. Weber, M. Hayashi, and S. H. Lin, Computational aspects of Franck-Condon overlap intervals. In Pandalai, S. G., ed., *Recent Research Developments in Physical Chemistry*, Vol. 8, Transworld Research Network (2005).

initially obtaining

$$I_4 = \sum_{\mu=0}^{\min(m_1,m_2)} 2^\mu \mu! \binom{m_1}{\mu}\binom{m_2}{\mu} \sum_{\nu=0}^{\min(m_3,m_4)} 2^\nu \nu! \binom{m_3}{\nu}\binom{m_4}{\nu}$$

$$\times \int_{-\infty}^{\infty} e^{-x^2} H_{m_1+m_2-2\mu}(x) H_{m_3+m_4-2\nu}(x)dx. \tag{18.38}$$

Invoking the orthogonality of the H_m with the weighting factor shown, the integral in Eq. (18.38) can be evaluated, yielding

$$\int_{-\infty}^{\infty} e^{-x^2} H_{m_1+m_2-2\mu}(x) H_{m_3+m_4-2\nu}(x)dx$$

$$= \sqrt{\pi}\, 2^{m_3+m_4-2\nu}(m_3+m_4-2\nu)!\, \delta_{m_1+m_2-2\mu,m_3+m_4-2\nu}. \tag{18.39}$$

The Kronecker delta in Eq. (18.39) limits the value of μ to the single value, if any, that satisfies

$$\mu = \frac{m_1+m_2-m_3-m_4}{2} + \nu, \tag{18.40}$$

so the double summation collapses to a single sum over ν. Moreover, when the powers of 2 in Eqs. (18.38) and (18.39) are combined, their resultant is 2^M, where

$$M = \frac{m_1+m_2+m_3+m_4}{2}. \tag{18.41}$$

We now rewrite Eq. (18.38), removing the μ summation and assigning μ the value from Eq. (18.40), writing the binomial coefficients in terms of their constituent factorials, and introducing M wherever it will result in simplification. We reach

$$I_4 = \sum_\nu \frac{\sqrt{\pi}\, 2^M\, (m_3+m_4-2\nu)!\, m_1!\, m_2!\, m_3!\, m_4!}{(M-m_3-m_4+\nu)!\,(M-m_1-\nu)!\,(M-m_2-\nu)!\,(m_3-\nu)!\,(m_4-\nu)!\,\nu!}. \tag{18.42}$$

This formula for I_4 will only be valid when the sum of the m_i is even, equivalent to the requirement that M (and therefore also μ) be integral. If the sum of the m_i is odd, then I_4 will have an odd integrand and will vanish by symmetry. The summation in Eq. (18.42) will be over the nonnegative integral values of ν for which none of the factorials in the denominator of that summation has a negative argument. Note that there will be no value of ν that satisfies this condition if $m_1 > m_2+m_3+m_4$, because $M-m_1$ will then be negative, and then $I_4 = 0$. Thus we have a generalization of the triangle condition that applied to the threefold Hermite formula: If the largest of the m_i is greater than the sum of the others, the H_m of smaller m cannot combine to yield a Hermite polynomial of sufficiently large index to avoid an orthogonality zero.

Further examination of the factorials in the denominator of Eq. (18.42) reveals that the lower limit of the summation will (if $m_1 \le m_2+m_3+m_4$) always be $\nu = 0$; note that

$M - m_3 - m_4$ will always be nonnegative. The upper limit of the summation will be the smaller of m_4 and $M - m_1$. ■

The Hermite polynomial product formula can also be applied to products of Hermite polynomials with a different exponential weighting function than in the examples we have presented. To evaluate such integrals we use the generalized product formula in conjunction with the integral (see Gradshteyn and Ryzhik, p. 803, in the Additional Readings),

$$\int_{-\infty}^{\infty} e^{-a^2x^2} H_m(x)H_n(x)dx = \frac{2^{m+n}}{a^{m+n+1}}(1-a^2)^{(m+n)/2}\Gamma\left(\frac{m+n+1}{2}\right)$$

$$\times \sum_{\nu=0}^{\min(m,n)} \frac{(-m)_\nu(-n)_\nu}{\nu!\left(\frac{1-m-n}{2}\right)_\nu}\left(\frac{a^2}{2(a^2-1)}\right)^\nu, \qquad (18.43)$$

instead of the standard orthogonality integral for the product of two Hermite polynomials. The quantity $(-m)_\nu$ is a Pochhammer symbol, and causes the ν summation in Eq. (18.43) to be a finite sum. The summation can also be identified as a hypergeometric function; see Exercise 18.5.11. The process we have sketched yields a result that is similar to I_m but somewhat more complicated. We omit details.

The oscillator potential has also been employed extensively in calculations of nuclear structure (nuclear shell model), as well as in quark models of hadrons and the nuclear force.

Exercises

18.2.1 Prove that $\left(2x - \frac{d}{dx}\right)^n 1 = H_n(x)$.

Hint. Check out the cases $n = 0$ and $n = 1$ and then use mathematical induction (Section 1.4).

18.2.2 Show that $\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)dx = 0$ for m an integer, $0 \le m \le n-1$.

18.2.3 The transition probability between two oscillator states m and n depends on

$$\int_{-\infty}^{\infty} xe^{-x^2} H_n(x)H_m(x)dx.$$

Show that this integral equals $\pi^{1/2}2^{n-1}n!\,\delta_{m,n-1} + \pi^{1/2}2^n(n+1)!\,\delta_{m,n+1}$. This result shows that such transitions can occur only between states of adjacent energy levels, $m = n \pm 1$.

Hint. Multiply the generating function, Eq. (18.2), by itself using two different sets of variables (x, s) and (x, t). Alternatively, the factor x may be eliminated by the recurrence relation, Eq. (18.3).

18.2.4 Show that $\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_n(x) dx = \pi^{1/2} 2^n n! \left(n + \frac{1}{2}\right)$.

This integral occurs in the calculation of the mean-square displacement of our quantum oscillator.

Hint. Use the recurrence relation, Eq. (18.3), and the orthogonality integral.

18.2.5 Evaluate

$$\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_m(x) dx$$

in terms of n and m and appropriate Kronecker delta functions.

ANS. $2^{n-1}\pi^{1/2}(2n+1)n!\,\delta_{nm} + 2^n \pi^{1/2}(n+2)!\,\delta_{n+2,m} + 2^{n-2}\pi^{1/2} n!\,\delta_{n-2,m}$.

18.2.6 Show that $\int_{-\infty}^{\infty} x^r e^{-x^2} H_n(x) H_{n+p}(x) dx = \begin{cases} 0, & p > r \\ 2^n \pi^{1/2}(n+r)!, & p = r, \end{cases}$

with n, p, and r nonnegative integers.

Hint. Use the recurrence relation, Eq. (18.3), p times.

18.2.7 With $\psi_n(x) = e^{-x^2/2} \dfrac{H_n(x)}{(2^n n!\, \pi^{1/2})^{1/2}}$, verify that

$$a\psi_n(x) = \frac{x - ip}{\sqrt{2}} = \frac{1}{\sqrt{2}}\left(x + \frac{d}{dx}\right)\psi_n(x) = n^{1/2}\psi_{n-1}(x),$$

$$a^\dagger \psi_n(x) = \frac{x + ip}{\sqrt{2}} = \frac{1}{\sqrt{2}}\left(x - \frac{d}{dx}\right)\psi_n(x) = (n+1)^{1/2}\psi_{n+1}(x).$$

Note. The usual quantum mechanical operator approach establishes these raising and lowering properties before the form of $\psi_n(x)$ is known.

18.2.8 (a) Verify the operator identity

$$x + ip = x - \frac{d}{dx} = -\exp\left[\frac{x^2}{2}\right]\frac{d}{dx}\exp\left[-\frac{x^2}{2}\right].$$

(b) The normalized simple harmonic oscillator wave function is

$$\psi_n(x) = (\pi^{1/2} 2^n n!)^{-1/2} \exp\left[-\frac{x^2}{2}\right] H_n(x).$$

Show that this may be written as

$$\psi_n(x) = (\pi^{1/2} 2^n n!)^{-1/2} \left(x - \frac{d}{dx}\right)^n \exp\left[-\frac{x^2}{2}\right].$$

Note. This corresponds to an n-fold application of the raising operator of Exercise 18.2.7.

18.3 Laguerre Functions

Rodrigues Formula and Generating Function

Let's start from the Laguerre ODE,

$$xy''(x) + (1-x)y'(x) + ny(x) = 0. \tag{18.44}$$

This ODE is not self-adjoint, but the weighting factor needed to make it self-adjoint can be computed from the usual formula,

$$w(x) = \frac{1}{x}\,\exp\left[\int \frac{1-x}{x}\,dx\right] = \frac{1}{x}\,\exp(\ln x - x) = e^{-x}. \tag{18.45}$$

Given $w(x)$, we may now use the method developed in Section 12.1 to obtain a Rodrigues formula and generating function for the Laguerre polynomials. Letting $L_n(x)$ denote the nth Laguerre polynomial, the Rodrigues formula is (apart from a scale factor) given by Eq. (12.9):

$$L_n(x) = \frac{1}{w(x)}\left(\frac{d}{dx}\right)^n \big(w(x)p(x)^n\big),$$

where $p(x)$ is the coefficient of y'' in the ODE. Inserting the expressions for $w(x)$ and $p(x)$, and inserting a factor $1/n!$ to bring the Laguerre polynomials to their conventional scaling, the Rodrigues formula takes the more complete and explicit form,

$$L_n(x) = \frac{e^x}{n!}\left(\frac{d}{dx}\right)^n \big(x^n e^{-x}\big). \tag{18.46}$$

A generating function can now be written as a sum of contour integrals of the Schlaefli type, as in Eq. (12.25):

$$g(x,t) = \sum_{t=0}^{\infty} L_n(x)t^n = \frac{1}{w(x)}\sum_{n=0}^{\infty}\frac{c_n\,t^n\,n!}{2\pi i}\oint_C \frac{w(z)[p(z)]^n}{(z-x)^{n+1}}\,dz,$$

where the contour surrounds the point x and no other singularities. Specializing to our current problem, and noting that the coefficient c_n has the value $1/n!$, this formula becomes

$$g(x,t) = \frac{e^x}{2\pi i}\sum_{n=0}^{\infty}\oint_C \frac{e^{-z}\,(tz)^n}{(z-x)^{n+1}}\,dz = \frac{e^x}{2\pi i}\oint_C \frac{e^{-z}\,dz}{(z-x)}\sum_{n=0}^{\infty}\left(\frac{tz}{z-x}\right)^n. \tag{18.47}$$

We now recognize the n summation as a geometric series, so our generating function becomes

$$g(x,t) = \frac{e^x}{2\pi i}\oint_C \frac{e^{-z}\,dz}{z-x-tz}. \tag{18.48}$$

Our integrand has a simple pole at $z = x/(1-t)$, with residue $e^{-x/(1-t)}/(1-t)$, and $g(x,t)$ reduces to

$$g(x,t) = \frac{e^{x}\, e^{-x/(1-t)}}{1-t} = \frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} L_n(x) t^n, \tag{18.49}$$

the form given in Table 12.1.

Not all workers define Laguerre polynomials at the scale chosen here and represented by the specific formulas in Eq. (18.46) and (18.49). However, our choice is probably the most common, and is consistent with that in AMS-55 (see Abramowitz in Additional Readings).

Properties of Laguerre Polynomials

By differentiating the generating function in Eq. (18.45) with respect to x and t, we obtain recurrence relations for the Laguerre polynomials as follows. Using the product rule for differentiation we verify the identities

$$(1-t)^2 \frac{\partial g}{\partial t} = (1-x-t) g(x,t), \qquad (t-1)\frac{\partial g}{\partial x} = t g(x,t). \tag{18.50}$$

Writing the left-hand and right-hand sides of the first identity in terms of Laguerre polynomials using the expansion given in Eq. (18.49), we obtain

$$\sum_n \big[(n+1) L_{n+1}(x) - 2n L_n(x) + (n-1) L_{n-1}(x) \big] t^n$$
$$= \sum_n \big[(1-x) L_n(x) - L_{n-1}(x) \big] t^n.$$

Equating coefficients of z^n yields

$$(n+1) L_{n+1}(x) = (2n+1-x) L_n(x) - n L_{n-1}(x). \tag{18.51}$$

To get the second recursion relation we use both identities of Eqs. (18.50) to verify a third identity,

$$x\frac{\partial g}{\partial x} = t\frac{\partial g}{\partial t} - t\frac{\partial (tg)}{\partial t},$$

which, when written similarly in terms of Laguerre polynomials, is seen to be equivalent to

$$x L_n'(x) = n L_n(x) - n L_{n-1}(x). \tag{18.52}$$

To use these recurrence formulas we need starting values. From the Rodrigues formula, we easily find $L_0(x) = 1$ and $L_1(x) = 1 - x$. Applying Eq. (18.51) we continue to $L_n(x)$ with $n > 1$, obtaining the results given in Table 18.2. The first three Laguerre polynomials are plotted in Fig. 18.3.

Table 18.2 Laguerre Polynomials

$L_0(x) = 1$
$L_1(x) = -x + 1$
$2!\,L_2(x) = x^2 - 4x + 2$
$3!\,L_3(x) = -x^3 + 9x^2 - 18x + 6$
$4!\,L_4(x) = x^4 - 16x^3 + 72x^2 - 96x + 24$
$5!\,L_5(x) = -x^5 + 25x^4 - 200x^3 + 600x^2 - 600x + 120$
$6!\,L_6(x) = x^6 - 36x^5 + 450x^4 - 2400x^3 + 5400x^2 - 4320x + 720$

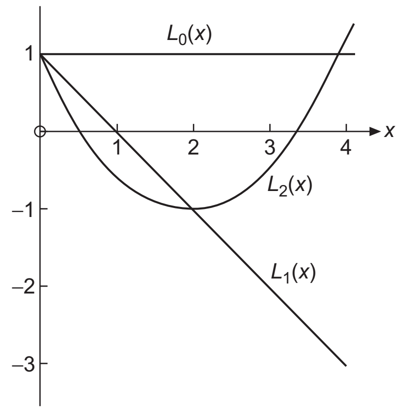

FIGURE 18.3 Laguerre polynomials.

From the recurrence relations or the Rodrigues formula, we find the the power series expansion of $L_n(x)$:

$$L_n(x) = \frac{(-1)^n}{n!}\left[x^n - \frac{n^2}{1!}x^{n-1} + \frac{n^2(n-1)^2}{2!}x^{n-2} - \cdots + (-1)^n n!\right]$$

$$= \sum_{m=0}^{n} \frac{(-1)^m n!\, x^m}{(n-m)!\, m!\, m!} = \sum_{s=0}^{n} \frac{(-1)^{n-s} n!\, x^{n-s}}{(n-s)!\,(n-s)!\, s!}. \tag{18.53}$$

Also, from Eq. (18.49) we find

$$g(0,t) = \frac{1}{1-t} = \sum_{n=0}^{\infty} t^n = \sum_{n=0}^{\infty} L_n(0) t^n,$$

which shows that at $x = 0$ the Laguerre polynomials have the special value

$$L_n(0) = 1. \tag{18.54}$$

The form of the generating function, that of Laguerre's ODE, and Table 18.2 all show that the Laguerre polynomials have neither odd nor even symmetry under the parity transformation $x \to -x$.

As we already observed at the beginning of this section, the Laguerre ODE is not self-adjoint but can be made so by appending the weighting factor e^{-x}. Noting also that with this weighting factor, the Laguerre polynomials satisfy Sturm-Liouville boundary conditions at $x = 0$ and $x = \infty$, we see that the $L_n(x)$ must satisfy an orthogonality condition of the form

$$\int_0^\infty e^{-x} L_m(x) L_n(x) dx = \delta_{mn}. \tag{18.55}$$

Equation (18.55) indicates that for this interval and weighting factor the Laguerre polynomials are normalized. Proof is the topic of Exercise 18.3.3.

It is sometimes convenient to define orthogonalized Laguerre functions (with unit weighting factor) by

$$\varphi_n(x) = e^{-x/2} L_n(x). \tag{18.56}$$

Our new orthonormal functions, $\varphi_n(x)$, satisfy the self-adjoint ODE

$$x\varphi_n''(x) + \varphi_n'(x) + \left(n + \frac{1}{2} - \frac{x}{4}\right)\varphi_n(x) = 0, \tag{18.57}$$

and are eigenfunctions of a Sturm-Liouville system on the range $(0 \le x < \infty)$.

Associated Laguerre Polynomials

In many applications, particularly in quantum mechanics, we need the associated Laguerre polynomials defined by[4]

$$L_n^k(x) = (-1)^k \frac{d^k}{dx^k} L_{n+k}(x). \tag{18.58}$$

By differentiating the power series for $L_n(x)$ given in Eq. (18.53) (compare Table 18.2), we can get the explicit forms shown in Table 18.3. In general,

$$L_n^k(x) = \sum_{m=0}^{n} (-1)^m \frac{(n+k)!}{(n-m)!\,(k+m)!\,m!} x^m, \quad k \ge 0. \tag{18.59}$$

One of the present authors[5] has recently found a new generating function for the associated Laguerre polynomials with the remarkably simple form

$$g_l(x,t) = e^{-tx}(1+t)^l = \sum_{n=0}^{\infty} L_n^{l-n}(x) t^n. \tag{18.60}$$

[4]Some authors use $\mathfrak{L}_{n+k}^k(x) = (d^k/dx^k)[L_{n+k}(x)]$. Hence our $L_n^k(x) = (-1)^k \mathfrak{L}_{n+k}^k(x)$.

[5]H. J. Weber, Connections between real polynomial solutions of hypergeometric-type differential equations with Rodrigues formula, *Cent. Eur. J. Math.* **5**: 415–427 (2007).

Table 18.3 Associated Laguerre Polynomials

$L_0^k = 1$
$1!\,L_1^k = -x + (k+1)$
$2!\,L_2^k = x^2 - 2(k+2)x + (k+1)_2$
$3!\,L_3^k = -x^3 + 3(k+3)x^2 - 3(k+2)_2 x + (k+1)_3$
$4!\,L_4^k = x^4 - 4(k+4)x^3 + 6(k+3)_2 - 4(k+2)_3 + (k+1)_4$
$5!\,L_5^k = -x^5 + 5(k+5)x^4 - 10(k+4)_2 x^3 + 10(k+3)_3 x^2 - 5(k+2)_4 x + (k+1)_5$
$6!\,L_6^k = x^6 - 6(k+6)x^5 + 15(k+5)_2 x^4 - 20(k+4)_3 x^3 + 15(k+3)_4 x^2$ $-6(k+2)_5 x + (k+1)_6$
$7!\,L_7^k = -x^7 + 7(k+7)x^6 - 21(k+6)_2 x^5 + 35(k+5)_3 x^4 - 35(k+4)_4 x^3$ $+21(k+3)_5 x^2 - 7(k+2)_6 x + (k+1)_7$

The notations $(k+n)_m$ are Pochhammer symbols, defined in Eq. (1.72).

Rather than deriving this formula, we verify it by showing that it produces the defining relation for the L_n^k, Eq. (18.58), and is consistent with the previously presented formulas for the ordinary Laguerre polynomials (i.e., the L_n^k with $k = 0$).

If we multiply both members of Eq. (18.60) by $1 - t$, the coefficients of t^n yield the recurrence formula

$$L_n^{l-n} + L_{n-1}^{l-n+1} = L_n^{l-n+1}, \quad \text{or} \quad L_n^k - L_n^{k+1} = -L_{n-1}^{k+1}. \tag{18.61}$$

On the other hand, differentiation of Eq. (18.60) with respect to x and writing

$$\frac{\partial g_l(x,t)}{\partial x} = \sum_n \frac{dL_n^{l-n}(x)}{dx}\, t^n = -te^{-tx}(1+t)^l = e^{-tx}(1+t)^l - e^{-tx}(1+t)^{l+1},$$

the coefficients of t^n yield a formula for $dL_n^{l-n}(x)/dx$, namely (with $k = l - n$)

$$\frac{dL_n^k(x)}{dx} = L_n^k(x) - L_n^{k+1}(x), \tag{18.62}$$

and substituting the result from Eq. (18.61), we reach

$$\frac{dL_n^k(x)}{dx} = -L_{n-1}^{k+1}, \tag{18.63}$$

thereby confirming that our generating function yields Eq. (18.58).

The verification that our generating function is correct is now completed by using it to find $L_n^0(x)$, which is the coefficient of t^n in $e^{-tx}(1+t)^n$. Using the binomial expansion of $(1+t)^n$ and the Maclaurin series for the exponential, we get

$$L_n^0(x) = \sum_{m=0}^{n} \binom{n}{n-m} \frac{(-x)^m}{m!} = \sum_{m=0}^{n} \frac{(-1)^m n!}{(n-m)!\,m!\,m!}\, x^m,$$

in agreement with Eq. (18.53).

We can also confirm the series expansion given as Eq. (18.59) for L_n^k. It is the coefficient of t^n in $e^{-tx}(1+t)^{k+n}$, obtained in a manner similar to the procedure we just carried out for L_n^0.

The generating function provides convenient routes to other properties of the associated Laguerre polynomials. Special values for $x=0$ can be obtained from

$$\sum_n L_n^{l-n}(0)t^n = (1+t)^l = \sum_{n=0}^{l} \binom{l}{n} t^n.$$

We therefore have

$$L_n^k(0) = \binom{n+k}{n}. \tag{18.64}$$

A formula for recurrence in the index n of $L_n^k(x)$ can be obtained by differentiating the generating function formula with respect to t. Doing so, from the coefficient of t^n and setting $l = k+n$,

$$(n+1)L_{n+1}^{k-1}(x) = (k+n)L_n^{k-1}(x) - xL_n^k(x). \tag{18.65}$$

Using Eq. (18.61) to raise the upper index in the two terms for which it is $k-1$, we find after collecting similar terms,

$$(n+1)L_{n+1}^k(x) - (2n+k+1-x)L_n^k(x) + (n+k)L_{n-1}^k(x) = 0, \tag{18.66}$$

a lower-index recurrence formula.

Finally, returning to Eq. (18.65), differentiating it once with respect to x, and identifying $\left[L_{n+1}^{k-1}\right]' = -L_n^k$, we get

$$(n+k)\left[L_n^{k-1}\right]' = x\left[L_n^k\right]' + L_n^k - (n+1)L_n^k = x\left[L_n^k\right]' - nL_n^k. \tag{18.67}$$

A second differentiation brings us to

$$x\left[L_n^k\right]'' + (1-n)\left[L_n^k\right]' = (n+k)\left[L_n^{k-1}\right]'' = (n+k)\left[L_n^{k-1}\right]' - (n-k)\left[L_n^k\right]', \tag{18.68}$$

where the final member of Eq. (18.68) was the result of substituting the derivative of Eq. (18.62) with $k \to k-1$. Using Eq. (18.67) to replace $(n+k)\left[L_n^{k-1}\right]'$ by a form in which the upper index is k, we reach an ODE for L_n^k:

$$x\frac{d^2L_n^k(x)}{dx^2} + (k+1-x)\frac{dL_n^k(x)}{dx} + nL_n^k(x) = 0. \tag{18.69}$$

This ODE is known as the **associated Laguerre equation**. When associated Laguerre polynomials appear in a physical problem it is usually because that physical problem involves Eq. (18.69). The most important application is their use to describe the bound states of the hydrogen atom, which are derived in upcoming Example 18.3.1.

The associated Laguerre equation, Eq. (18.69), is not self-adjoint, but the weighting function needed to bring it to self-adjoint form (for upper index k) can be found in the usual way:

$$w_k(x) = \frac{1}{x} \exp\left[\int \frac{k+1-x}{x}\,dx\right] = x^k e^{-x}. \tag{18.70}$$

When we also note that Sturm-Liouville boundary conditions are satisfied at $x = 0$ and $x = \infty$, we see that the associated Laguerre polynomials are orthogonal according to the equation

$$\int_0^\infty e^{-x} x^k L_n^k(x) L_m^k(x) dx = \frac{(n+k)!}{n!}\,\delta_{mn}. \tag{18.71}$$

The value of the integral in Eq. (18.71) for $m = n$ can be established using the generating function, Eq. (18.58). Doing so is left as an exercise.

Equation (18.71) shows the same orthogonality interval $(0, \infty)$ as that for the Laguerre polynomials, but with a different weighting function for each k. We see that for each k the associated Laguerre polynomials define a new set of orthogonal polynomials.

A Rodrigues representation of the associated Laguerre polynomials is useful and can be found in various ways. A fairly direct approach is simply to use Eq. (12.9) with $p(x) = x$, the coefficient of the second-derivative term in Eq. (18.69) and the value of $w_k(x)$ given in Eq. (18.70). The result is

$$L_n^k(x) = \frac{e^x x^{-k}}{n!} \frac{d^n}{dx^n} (e^{-x} x^{n+k}). \tag{18.72}$$

Note that this and all our earlier formulas involving the $L_n^k(x)$ reduce properly to corresponding expressions involving $L_n(x)$ when $k = 0$.

By letting $\psi_n^k(x) = e^{-x/2} x^{k/2} L_n^k(x)$, we find that $\psi_n^k(x)$ satisfies the self-adjoint ODE,

$$x\,\frac{d^2\psi_n^k(x)}{dx^2} + \frac{d\psi_n^k(x)}{dx} + \left(-\frac{x}{4} + \frac{2n+k+1}{2} - \frac{k^2}{4x}\right)\psi_n^k(x) = 0. \tag{18.73}$$

The $\psi_n^k(x)$ are sometimes called **Laguerre functions**. Equation (18.57) is the special case $k = 0$ of Eq. (18.73).

A further useful form is given by defining[6]

$$\Phi_n^k(x) = e^{-x/2} x^{(k+1)/2} L_n^k(x). \tag{18.74}$$

Substitution into the associated Laguerre equation yields

$$\frac{d^2\Phi_n^k(x)}{dx^2} + \left(-\frac{1}{4} + \frac{2n+k+1}{2x} - \frac{k^2-1}{4x^2}\right)\Phi_n^k(x) = 0. \tag{18.75}$$

The $\Phi_n^k(x)$ are orthogonal with weighting function x^{-1}.

The associated Laguerre ODE, Eq. (18.69), has solutions even if n is not an integer, but they are then not polynomials and diverge proportionally to xke^x as $x \to \infty$. This fact is useful in the following example.

[6] This corresponds to modifying the function ψ in Eq. (18.73) to eliminate the first derivative.

Example 18.3.1 The Hydrogen Atom

The most important application of the Laguerre polynomials is in the solution of the Schrödinger equation for the hydrogen-like atom (H, He^+, Li^{2+}, etc.). For a system consisting of a nucleus of charge Ze fixed at the origin and one electron whose distribution is described by a wave function ψ, this equation is

$$-\frac{\hbar^2}{2m}\nabla^2\psi - \frac{Ze^2}{4\pi\epsilon_0 r}\psi = E\psi, \tag{18.76}$$

in which $Z = 1$ for hydrogen, $Z = 2$ for He^+, and so on. Separating variables in spherical polar coordinates and recognizing that the angular part of the solution to this equation must be a spherical harmonic, we set $\psi(\mathbf{r}) = R(r)Y_L^M(\theta,\varphi)$ with $R(r)$ satisfying the ODE

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{Ze^2}{4\pi\epsilon_0 r}R + \frac{\hbar^2}{2m}\frac{L(L+1)}{r^2}R = E. \tag{18.77}$$

For bound states, $R \to 0$ as $r \to \infty$, and it can be shown that these conditions can only be met if $E < 0$. In addition, R must be finite at $r = 0$. We do not consider unbound (continuum) states with positive energy. Only when the latter are included do hydrogenic wave functions form a complete set.

By use of the abbreviations (resulting from rescaling r to the dimensionless radial variable ρ)

$$\alpha = \left[-\frac{8mE}{\hbar^2}\right]^{1/2}, \quad \rho = \alpha r, \quad \lambda = \frac{mZe^2}{2\pi\epsilon_0\alpha\hbar^2}, \quad \chi(\rho) = R(r), \tag{18.78}$$

Eq. (13.85) becomes

$$\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{d\chi(\rho)}{d\rho}\right) + \left(\frac{\lambda}{\rho} - \frac{1}{4} - \frac{L(L+1)}{\rho^2}\right)\chi(\rho) = 0. \tag{18.79}$$

For our present purposes, it is useful to rewrite the first term of Eq. (18.79) using the identity

$$\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{d\chi}{d\rho}\right) = \frac{1}{\rho}\frac{d^2}{d\rho^2}(\rho\chi)$$

and then multiply the resulting equation by ρ, reaching

$$\frac{d}{d\rho^2}(\rho\chi) + \left(\frac{\lambda}{\rho} - \frac{1}{4} - \frac{L(L+1)}{\rho^2}\right)(\rho\chi) = 0. \tag{18.80}$$

A comparison with Eq. (18.75) for $\Phi_n^k(x)$ shows that Eq. (18.80) is satisfied by

$$\rho\chi(\rho) = e^{-\rho/2}\rho^{L+1}L_{\lambda-L-1}^{2L+1}(\rho), \tag{18.81}$$

where k and n of Eq. (18.75) have been, respectively, replaced by $2L+1$ and $\lambda - L - 1$.

The parameter λ must be restricted to values such that $\lambda - L - 1$ is both integral and nonnegative. If this requirement is violated, $L_{\lambda-L-1}^{2L+1}$ will diverge too rapidly to permit $\rho\chi(\rho)$ to go to zero at large r, which is required for a bound-state electron distribution.

Since we already know that L, a spherical harmonic index, must be integral and nonnegative, we see that the possible values of λ are integers n at least as large as $L+1$.[7]

This restriction on λ, imposed by our boundary condition, has the effect of quantizing the energy. Inserting $\lambda = n$, the definitions in Eqs. (18.78) lead to

$$E_n = -\frac{Z^2 m}{2n^2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2. \tag{18.82}$$

Since our Schrödinger equation implicitly set the potential energy to zero when the electron is at an infinite separation from the nucleus, the negative sign reflects the fact that we are dealing here with bound states in which the electron cannot escape to infinity. The other quantities introduced in Eq. (18.78) can also be expressed in terms of n:

$$\alpha = \frac{me^2}{2\pi\epsilon_0\hbar^2}\frac{Z}{n} = \frac{2Z}{na_0}, \quad \rho = \frac{2Z}{na_0}r, \quad \text{with } a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}. \tag{18.83}$$

The quantity a_0, of dimension length, is known as the **Bohr radius**, and its appearance as a scale factor causes the potential energy (for $n = 1$, the smallest possible value) to have an average value corresponding to this electron-nuclear separation.

Summarizing, the final normalized hydrogen wave function is

$$\psi_{nLM}(r,\theta,\varphi) = \left[\left(\frac{2Z}{na_0}\right)^3 \frac{(n-L-1)!}{2n(n+L)!}\right]^{1/2} e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r) Y_L^M(\theta,\varphi). \tag{18.84}$$

Note that the energy corresponding to ψ_{nLM} depends only on n, which is called the **principal quantum number** of this system. Note also that if n is assigned a specific integral value, the condition on λ requires that $L \le n-1$, thereby explaining the well-known pattern of possible hydrogenic energy states: If $n = 1$, L can only be zero; for $n = 2$, we can have $L = 0$ or $L = 1$, etc. ■

Exercises

18.3.1 Show with the aid of the Leibniz formula that the series expansion of $L_n(x)$, Eq. (18.53), follows from the Rodrigues representation, Eq. (18.72).

18.3.2 (a) Using the explicit series form, Eq. (18.53), show that

$$L_n'(0) = -n, \quad L_n''(0) = \tfrac{1}{2}n(n-1).$$

(b) Repeat without using the explicit series form of $L_n(x)$.

18.3.3 Derive the normalization relation, Eq. (18.71) for the associated Laguerre polynomials, thereby also confirming Eq. (18.55) for the L_n.

[7] This is the conventional notation for λ. It is not the same n as the index n in $\Phi_n^k(x)$.

18.3.4 Expand x^r in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to r (or to ∞ if r is not an integer).

Hint. The Rodrigues form of $L_n^k(x)$ will be useful.

$$\textit{ANS.} \quad x^r = (r+k)!\,r! \sum_{n=0}^{r} \frac{(-1)^n L_n^k(x)}{(n+k)!(r-n)!}, \qquad 0 \le x < \infty.$$

18.3.5 Expand e^{-ax} in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to ∞.

(a) Evaluate directly the coefficients in your assumed expansion.
(b) Develop the desired expansion from the generating function.

$$\textit{ANS.} \quad e^{-ax} = \frac{1}{(1+a)^{1+k}} \sum_{n=0}^{\infty} \left(\frac{a}{1+a}\right)^n L_n^k(x), \qquad 0 \le x < \infty.$$

18.3.6 Show that $\displaystyle\int_0^\infty e^{-x} x^{k+1} L_n^k(x) L_n^k(x)\,dx = \frac{(n+k)!}{n!}(2n+k+1).$

Hint. Note that $xL_n^k = (2n+k+1)L_n^k - (n+k)L_{n-1}^k - (n+1)L_{n+1}^k.$

18.3.7 Assume that a particular problem in quantum mechanics has led to the ODE

$$\frac{d^2y}{dx^2} - \left[\frac{k^2-1}{4x^2} - \frac{2n+k+1}{2x} + \frac{1}{4}\right] y = 0$$

for nonnegative integers n, k. Write $y(x)$ as $y(x) = A(x)B(x)C(x)$, with the requirement that

(a) $A(x)$ be a **negative** exponential giving the required asymptotic behavior of $y(x)$, and
(b) $B(x)$ be a **positive** power of x giving the behavior of $y(x)$ for $0 \le x \ll 1$.

Determine $A(x)$ and $B(x)$. Find the relation between $C(x)$ and the associated Laguerre polynomial.

$$\textit{ANS.} \quad A(x) = e^{-x/2}, \quad B(x) = x^{(k+1)/2}, \quad C(x) = L_n^k(x).$$

18.3.8 From Eq. (18.84) the normalized radial part of the hydrogenic wave function is

$$R_{nL}(r) = \left[\alpha^3 \frac{(n-L-1)!}{2n(n+L)!}\right]^{1/2} e^{-\alpha r}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r),$$

in which $\alpha = 2Z/na_0 = 2Zme^2/4\pi\epsilon_0\hbar^2$. Evaluate

$$\text{(a) } \langle r \rangle = \int_0^\infty r R_{nL}(\alpha r) R_{nL}(\alpha r) r^2\,dr,$$

$$\text{(b) } \langle r^{-1}\rangle = \int_0^\infty r^{-1} R_{nL}(\alpha r) R_{nL}(\alpha r) r^2\, dr.$$

The quantity $\langle r\rangle$ is the average displacement of the electron from the nucleus, whereas $\langle r^{-1}\rangle$ is the average of the reciprocal displacement.

$$\textit{ANS.} \quad \langle r\rangle = \frac{a_0}{2}\left[3n^2 - L(L+1)\right], \qquad \langle r^{-1}\rangle = \frac{1}{n^2 a_0}.$$

18.3.9 Derive a recurrence formula for the hydrogen wave function expectation values:

$$\frac{s+2}{n^2}\langle r^{s+1}\rangle - (2s+3)a_0\langle r^s\rangle + \frac{s+1}{4}\left[(2L+1)^2 - (s+1)^2\right]a_0^2\langle r^{s-1}\rangle = 0,$$

with $s \geq -2L-1$.

Hint. Transform Eq. (18.80) into a form analogous to Eq. (18.73). Multiply by $\rho^{s+2}u' - c\rho^{s+1}u$, with $u = \rho\Phi$. Adjust c to cancel terms that do not yield expectation values.

18.3.10 Show that $\displaystyle\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)dx = \sqrt{\pi}\, n!\, P_n(y)$, where P_n is a Legendre polynomial.

18.4 Chebyshev Polynomials

The generating function for the Legendre polynomials can be generalized to the following form:

$$\frac{1}{(1-2xt+t^2)^\alpha} = \sum_{n=0}^\infty C_n^{(\alpha)}(x)t^n. \tag{18.85}$$

The coefficients $C_n^{(\alpha)}(x)$ are known as the **ultraspherical polynomials** (also called **Gegenbauer polynomials**). For $\alpha = 1/2$, we recover the Legendre polynomials; the special cases $\alpha = 0$ and $\alpha = 1$ yield two types of Chebyshev polynomials that are the subject of this section. The primary importance of the Chebyshev polynomials is in numerical analysis.

Type II Polynomials

With $\alpha = 1$ and $C_n^{(1)}(x)$ written as $U_n(x)$, Eq. (18.85) gives

$$\frac{1}{1-2xt+t^2} = \sum_{n=0}^\infty U_n(x)t^n, \quad |x| < 1, \quad |t| < 1. \tag{18.86}$$

These functions are called type II Chebyshev polynomials. Although these polynomials have few applications in mathematical physics, one unusual application is in the development of four-dimensional spherical harmonics used in angular momentum theory.

Type I Polynomials

With $\alpha = 0$ there is a difficulty. Indeed, our generating function reduces to the constant 1. We may avoid this problem by first differentiating Eq. (18.85) with respect to t. This yields

$$\frac{-\alpha(-2x+2t)}{(1-2xt+t^2)^{\alpha+1}} = \sum_{n=1}^{\infty} nC_n^{(\alpha)}(x)\,t^{n-1},$$

or

$$\frac{x-t}{(1-2xt+t^2)^{\alpha+1}} = \sum_{n=1}^{\infty} \frac{n}{2}\left[\frac{C_n^{(\alpha)}(x)}{\alpha}\right] t^{n-1}. \tag{18.87}$$

We define $C_n^{(0)}(x)$ as

$$C_n^{(0)}(x) = \lim_{\alpha\to 0} \frac{C_n^{(\alpha)}(x)}{\alpha}. \tag{18.88}$$

The purpose of differentiating with respect to t was to get α in the denominator and to create an indeterminate form. Now multiplying Eq. (18.87) by $2t$ and adding 1 in the form $(1-2xt+t^2)/(1-2xt+t^2)$, we obtain

$$\frac{1-t^2}{1-2xt+t^2} = 1 + 2\sum_{n=1}^{\infty} \frac{n}{2}\, C_n^{(0)}(x)t^n. \tag{18.89}$$

We define $T_n(x)$ as

$$T_n(x) = \begin{cases} 1, & n = 0, \\ \dfrac{n}{2}\, C_n^{(0)}(x), & n > 0. \end{cases} \tag{18.90}$$

Note the special treatment for $n = 0$. We will encounter a similar treatment of the $n = 0$ term when we study Fourier series in Chapter 19. Also, note that $C_n^{(0)}$ is the limit indicated in Eq. (18.88) and not a literal substitution of $\alpha = 0$ into the generating function series. With these new labels,

$$\frac{1-t^2}{1-2xt+t^2} = T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n, \quad |x| \le 1, \quad |t| < 1. \tag{18.91}$$

We call $T_n(x)$ the type I Chebyshev polynomials. Note that the notation and spelling of the name for these functions differ from reference to reference. Here we follow the usage of AMS-55 (Additional Readings).

Recurrence Relations

Differentiating the generating function, Eq. (18.91), with respect to t and multiplying by the denominator, $1 - 2xt + t^2$, we obtain

$$-t - (t - x)\left[T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n\right] = (1 - 2xt + t^2)\sum_{n=1}^{\infty} nT_n(x)t^{n-1}$$

$$= \sum_{n=1}^{\infty}\left[nT_n t^{n-1} - 2xnT_n t^n + nT_n t^{n+1}\right],$$

from which after several simplification steps we reach the recurrence relation

$$T_{n+1}(x) - 2xT_n(x) + T_{n-1}(x) = 0, \quad n > 0. \tag{18.92}$$

A similar treatment of Eq. (18.86) yields the corresponding recursion relation for U_n:

$$U_{n+1}(x) - 2xU_n(x) + U_{n-1}(x) = 0, \quad n > 0. \tag{18.93}$$

Using the generating functions directly for $n = 0$ and 1, and then applying these recurrence relations for the higher-order polynomials, we get Table 18.4. Plots of the T_n and U_n are presented in Figs. 18.4 and 18.5.

Differentiation of the generating functions for $T_n(x)$ and $U_n(x)$ with respect to the variable x leads to a variety of recurrence relations involving derivatives. For example, from Eq. (18.89) we thus obtain

$$(1 - 2xt + t^2)2\sum_{n=1}^{\infty} T_n'(x)t^n = 2t\left[T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n\right],$$

from which we extract the recursion formula

$$2T_n(x) = T_{n+1}'(x) - 2xT_n'(x) + T_{n-1}'(x). \tag{18.94}$$

Other useful recurrence formulas we can find in this way are

$$(1 - x^2)T_n'(x) = -nxT_n(x) + nT_{n-1}(x) \tag{18.95}$$

Table 18.4 Chebyshev Polynomials: Type I (Left), Type II (Right)

$T_0 = 1$	$U_0 = 1$
$T_1 = x$	$U_1 = 2x$
$T_2 = 2x^2 - 1$	$U_2 = 4x^2 - 1$
$T_3 = 4x^3 - 3x$	$U_3 = 8x^3 - 4x$
$T_4 = 8x^4 - 8x^2 + 1$	$U_4 = 16x^4 - 12x^2 + 1$
$T_5 = 16x^5 - 20x^3 + 5x$	$U_5 = 32x^5 - 32x^3 + 6x$
$T_6 = 32x^6 - 48x^4 + 18x^2 - 1$	$U_6 = 64x^6 - 80x^4 + 24x^2 - 1$

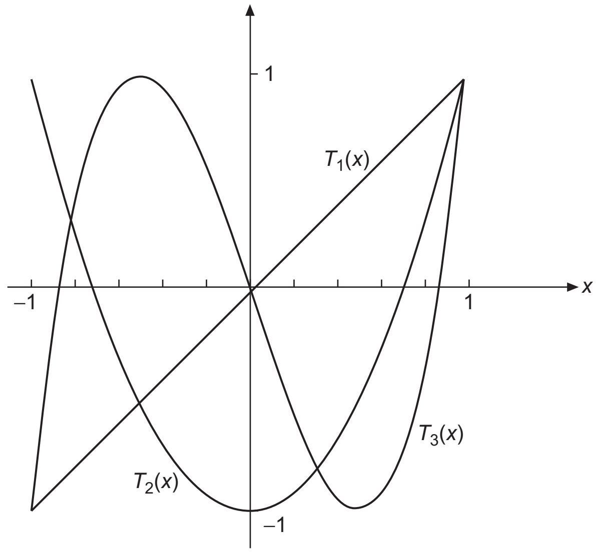

FIGURE 18.4 The Chebyshev polynomials T_1, T_2, and T_3.

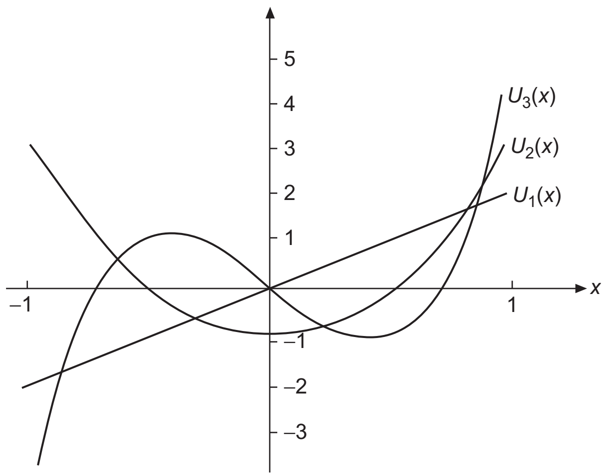

FIGURE 18.5 The Chebyshev polynomials U_1, U_2, and U_3.

and

$$(1 - x^2)U_n'(x) = -nxU_n(x) + (n+1)U_{n-1}(x). \tag{18.96}$$

Manipulating a variety of these formulas as in Section 15.1 for Legendre polynomials one can eliminate the index $n-1$ in favor of T_n'' and establish that $T_n(x)$, the Chebyshev polynomial type I, satisfies the ODE

$$(1-x^2)T_n''(x) - xT_n'(x) + n^2 T_n(x) = 0. \tag{18.97}$$

The Chebyshev polynomial of type II, $U_n(x)$, satisfies

$$(1-x^2)U_n''(x) - 3xU_n'(x) + n(n+2)U_n(x) = 0. \tag{18.98}$$

We could have defined the Chebyshev polynomials starting from these ODEs, but we chose instead a development based on generating functions.

Processes similar to those used for the Chebyshev polynomials can be applied to the general ultraspherical polynomials; the result is the **ultraspherical ODE**

$$(1-x^2)\frac{d^2}{dx^2}C_n^{(\alpha)}(x) - (2\alpha+1)x\frac{d}{dx}C_n^{(\alpha)}(x) + n(n+2\alpha)C_n^{(\alpha)}(x) = 0. \tag{18.99}$$

Special Values

Again, from the generating functions, we can obtain the special values of various polynomials:

$$\begin{gathered} T_n(1) = 1, \quad T_n(-1) = (-1)^n, \\ T_{2n}(0) = (-1)^n, \quad T_{2n+1}(0) = 0; \\ U_n(1) = n+1\,, \quad U_n(-1) = (-1)^n(n+1), \\ U_{2n}(0) = (-1)^n, \quad U_{2n+1}(0) = 0. \end{gathered} \tag{18.100}$$

Verification of Eq. (18.100) is left to the exercises.

The polynomials T_n and U_n satisfy parity relations that follow from their generating functions with the substitutions $t \to -t, x \to -x$, which leave them invariant; these are

$$T_n(x) = (-1)^n T_n(-x), \quad U_n(x) = (-1)^n U_n(-x). \tag{18.101}$$

Rodrigues representations of $T_n(x)$ and $U_n(x)$ are

$$T_n(x) = \frac{(-1)^n \pi^{1/2}(1-x^2)^{1/2}}{2^n \Gamma(n+\frac{1}{2})}\frac{d^n}{dx^n}\left[(1-x^2)^{n-1/2}\right] \tag{18.102}$$

and

$$U_n(x) = \frac{(-1)^n (n+1)\pi^{1/2}}{2^{n+1}\Gamma(n+\frac{3}{2})(1-x^2)^{1/2}}\frac{d^n}{dx^n}\left[(1-x^2)^{n+1/2}\right]. \tag{18.103}$$

Trigonometric Form

At this point in the development of the properties of the Chebyshev polynomials it is beneficial to change variables, replacing x by $\cos\theta$. With $x = \cos\theta$ and $d/dx = (-1/\sin\theta)(d/d\theta)$, we verify that

$$(1-x^2)\frac{d^2T_n}{dx^2} = \frac{d^2T_n}{d\theta^2} - \cot\theta\frac{dT_n}{d\theta}, \quad xT_n' = -\cot\theta\frac{dT_n}{d\theta}.$$

Adding these terms, Eq. (18.97) becomes

$$\frac{d^2T_n}{d\theta^2} + n^2T_n = 0, \tag{18.104}$$

the simple harmonic oscillator equation with solutions $\cos n\theta$ and $\sin n\theta$. The special values (boundary conditions at $x = 0$ and 1) identify

$$T_n = \cos n\theta = \cos(n \arccos x). \tag{18.105}$$

For $n \neq 0$ a second linearly independent solution of Eq. (18.104) is labeled

$$V_n = \sin n\theta = \sin(n \arccos x). \tag{18.106}$$

The corresponding solutions of the type II Chebyshev equation, Eq. (18.98), become

$$U_n = \frac{\sin(n+1)\theta}{\sin\theta}, \tag{18.107}$$

$$W_n = \frac{\cos(n+1)\theta}{\sin\theta}. \tag{18.108}$$

The two sets of solutions, type I and type II, are related by

$$V_n(x) = (1-x^2)^{1/2}U_{n-1}(x), \tag{18.109}$$

$$W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x). \tag{18.110}$$

As already seen from the generating functions, $T_n(x)$ and $U_n(x)$ are polynomials. Clearly, $V_n(x)$ and $W_n(x)$ are **not** polynomials. From

$$\begin{aligned} T_n(x) + iV_n(x) &= \cos n\theta + i\sin n\theta \\ &= (\cos\theta + i\sin\theta)^n = \left[x + i(1-x^2)^{1/2}\right]^n, \quad |x| \le 1 \end{aligned} \tag{18.111}$$

we can apply the binomial theorem to obtain expansions

$$T_n(x) = x^n - \binom{n}{2}x^{n-2}(1-x^2) + \binom{n}{4}x^{n-4}(1-x^2)^2 - \cdots \tag{18.112}$$

and, for $n > 0$

$$V_n(x) = \sqrt{1-x^2}\left[\binom{n}{1}x^{n-1} - \binom{n}{3}x^{n-3}(1-x^2) + \cdots\right]. \tag{18.113}$$

From the generating functions, or from the ODEs, power-series representations are

$$T_n(x) = \frac{n}{2}\sum_{m=0}^{[n/2]}(-1)^m \frac{(n-m-1)!}{m!\,(n-2m)!}\,(2x)^{n-2m} \tag{18.114}$$

for $n \geq 1$, with $[n/2]$ the integer part of $n/2$ and

$$U_n(x) = \sum_{m=0}^{[n/2]}(-1)^m \frac{(n-m)!}{m!\,(n-2m)!}\,(2x)^{n-2m}. \tag{18.115}$$

Application to Numerical Analysis

An important feature of the Chebyshev polynomials $T_n(x)$ with $n > 0$ is that as x is varied, they oscillate between the extreme values $T_n = +1$ and $T_n = -1$. This behavior is readily seen from Eq. (18.105) and is illustrated for T_{12} in Fig. 18.6. If a function is expanded in the T_n and the expansion is extended sufficiently that the contributions of successive T_n are decreasing rapidly, a good approximation to the truncation error will be proportional to the first T_n not included in the expansion. In this approximation, there will be negligible error at the n values of x where T_n is zero, and there will be maximum errors (all of the same magnitude but alternating in sign) at the extrema of T_n that fall between the zeros. In that sense, the errors satisfy a minimax principle, meaning that the maximum of the error has been minimized by distributing it evenly into the regions between the points of negligible error.

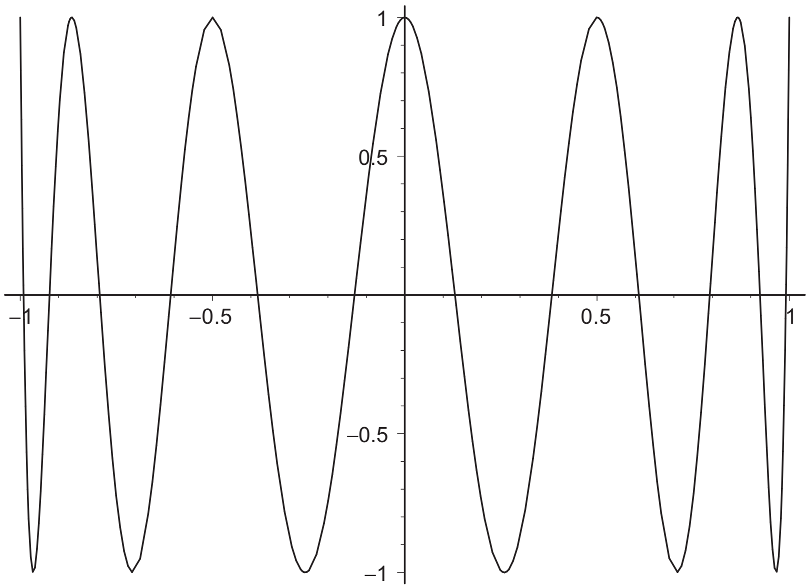

FIGURE 18.6 The Chebyshev polynomial T_{12}.

Example 18.4.1 Minimizing the Maximum Error

Figure 18.7 shows the errors in four-term expansions of e^x on the range $[-1, 1]$ carried out in various ways: (a) Maclaurin series, (b) Legendre expansion, and (c) Chebyshev expansion. The power series is optimum at the point $x = 0$ and the error increases with increasing values of $|x|$. The orthogonal expansions produce a fit over the region $[-1, 1]$, with the maximum errors occurring at $x = \pm 1$ and three intermediate values of x. However, the Legendre expansion has larger errors at ± 1 than it has at the interior points, while the Chebyshev expansion yields smaller errors at ± 1 (with a concomitant increase in the error at the other maxima) with the result that all the error maxima are comparable. This choice approximately minimizes the maximum error.

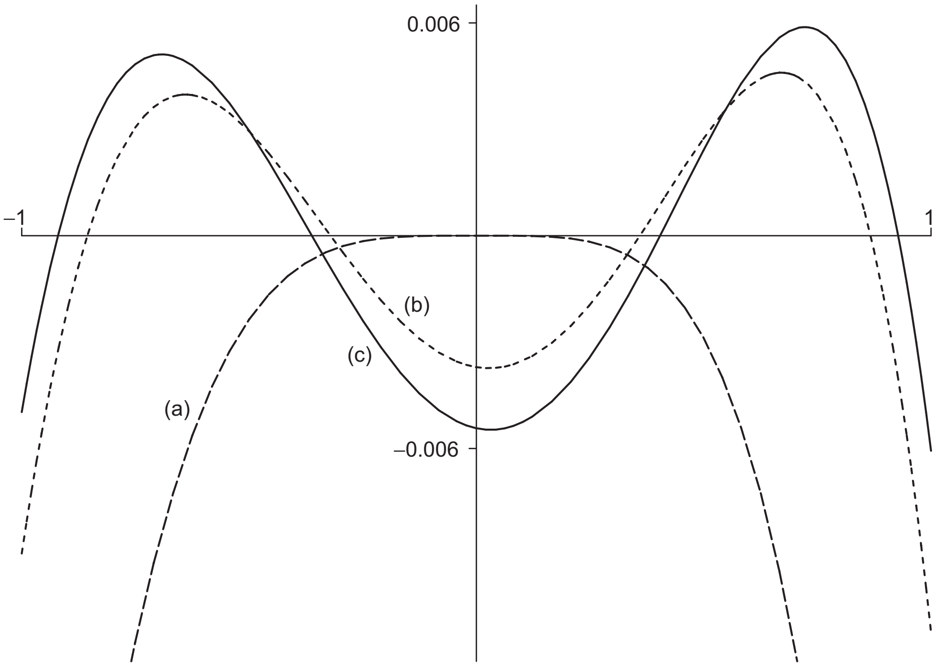

Figure 18.7 Error in four-term approximations to e^x: (a) Power series; (b) Legendre expansion; and (c) Chebyshev expansion.

■

Orthogonality

If Eq. (18.97) is put into self-adjoint form (Section 8.2), we obtain $w(x) = (1 - x^2)^{-1/2}$ as a weighting factor. For Eq. (18.98) the corresponding weighting factor is $(1 - x^2)^{+1/2}$.

The resulting orthogonality integrals,

$$\int_{-1}^{1} T_m(x)T_n(x)(1-x^2)^{-1/2}\,dx = \begin{cases} 0, & m \neq n, \\ \dfrac{\pi}{2}, & m = n \neq 0, \\ \pi, & m = n = 0, \end{cases} \tag{18.116}$$

$$\int_{-1}^{1} V_m(x)V_n(x)(1-x^2)^{-1/2}\,dx = \begin{cases} 0, & m \neq n, \\ \dfrac{\pi}{2}, & m = n \neq 0, \\ 0, & m = n = 0, \end{cases} \tag{18.117}$$

$$\int_{-1}^{1} U_m(x)U_n(x)(1-x^2)^{1/2}\,dx = \frac{\pi}{2}\,\delta_{mn}, \tag{18.118}$$

and

$$\int_{-1}^{1} W_m(x)W_n(x)(1-x^2)^{1/2}\,dx = \frac{\pi}{2}\,\delta_{mn}, \tag{18.119}$$

are a direct consequence of the Sturm-Liouville theory. The normalization values may best be obtained by making the substitution $x = \cos\theta$.

Exercises

18.4.1 By evaluating the generating function for special values of x, verify the special values

$$T_n(1) = 1, \quad T_n(-1) = (-1)^n, \quad T_{2n}(0) = (-1)^n, \quad T_{2n+1}(0) = 0.$$

18.4.2 By evaluating the generating function for special values of x, verify the special values

$$U_n(1) = n+1, \quad U_n(-1) = (-1)^n(n+1), \quad U_{2n}(0) = (-1)^n, \quad U_{2n+1}(0) = 0.$$

18.4.3 Another Chebyshev generating function is

$$\frac{1-xt}{1-2xt+t^2} = \sum_{n=0}^{\infty} X_n(x)t^n, \quad |t| < 1.$$

How is $X_n(x)$ related to $T_n(x)$ and $U_n(x)$?

18.4.4 Given

$$(1-x^2)U_n''(x) - 3xU_n'(x) + n(n+2)U_n(x) = 0,$$

show that $V_n(x)$, Eq. (18.106), satisfies

$$(1-x^2)V_n''(x) - xV_n'(x) + n^2V_n(x) = 0,$$

which is Chebyshev's equation.

18.4.5 Show that the Wronskian of $T_n(x)$ and $V_n(x)$ is given by

$$T_n(x)V_n'(x) - T_n'(x)V_n(x) = -\frac{n}{(1-x^2)^{1/2}}.$$

This verifies that T_n and V_n $(n \neq 0)$ are independent solutions of Eq. (18.97). Conversely, for $n = 0$, we do not have linear independence. What happens at $n = 0$? Where is the "second" solution?

18.4.6 Show that $W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x)$ is a solution of

$$(1-x^2)W_n''(x) - 3xW_n'(x) + n(n+2)W_n(x) = 0.$$

18.4.7 Evaluate the Wronskian of $U_n(x)$ and $W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x)$.

18.4.8 $V_n(x) = (1-x^2)^{1/2}U_{n-1}(x)$ is not defined for $n = 0$. Show that a second and independent solution of the Chebyshev differential equation for $T_n(x)$ $(n = 0)$ is $V_0(x) = \arccos x$ (or $\arcsin x$).

18.4.9 Show that $V_n(x)$ satisfies the same three-term recurrence relation as $T_n(x)$, Eq. (18.92).

18.4.10 Verify the series solutions for $T_n(x)$ and $U_n(x)$, Eqs. (18.114) and (18.115).

18.4.11 Transform the series form of $T_n(x)$, Eq. (18.114), into an **ascending** power series.

$$\textit{ANS.}\quad T_{2n}(x) = (-1)^n n \sum_{m=0}^{n} (-1)^m \frac{(n+m-1)!}{(n-m)!(2m)!}(2x)^{2m}, \quad n \geq 1,$$

$$T_{2n+1}(x) = \frac{2n+1}{2}\sum_{m=0}^{n} \frac{(-1)^{m+n}(n+m)!}{(n-m)!(2m+1)!}(2x)^{2m+1}.$$

18.4.12 Rewrite the series form of $U_n(x)$, Eq. (18.115), as an ascending power series.

$$\textit{ANS.}\quad U_{2n}(x) = (-1)^n \sum_{m=0}^{n} (-1)^m \frac{(n+m)!}{(n-m)!(2m)!}(2x)^{2m},$$

$$U_{2n+1}(x) = (-1)^n \sum_{m=0}^{n} (-1)^m \frac{(n+m+1)!}{(n-m)!(2m+1)!}(2x)^{2m+1}.$$

18.4.13 (a) From the differential equation for T_n (in self-adjoint form) show that

$$\int_{-1}^{1} \frac{dT_m(x)}{dx}\frac{dT_n(x)}{dx}(1-x^2)^{1/2}dx = 0, \qquad m \neq n.$$

(b) Confirm the preceding result by showing that

$$\frac{dT_n(x)}{dx} = nU_{n-1}(x).$$

18.4.14 The substitution $x = 2x' - 1$ converts $T_n(x)$ into the **shifted** Chebyshev polynomials $T_n^*(x')$. Verify that this produces the shifted polynomials shown in Table 18.5 and that

Table 18.5 Shifted Type I Chebyshev Polynomials

$T_0^* = 1$
$T_1^* = 2x - 1$
$T_2^* = 8x^2 - 8x + 1$
$T_3^* = 32x^3 - 48x^2 + 18x - 1$
$T_4^* = 128x^4 - 256x^3 + 160x^2 - 32x + 1$
$T_5^* = 512x^5 - 1280x^4 + 120x^3 - 400x^2 + 50x - 1$
$T_6^* = 2048x^6 - 6144x^5 + 6912x^4 - 3584x^3 + 840x^2 - 72x + 1$

they satisfy the orthonormality condition

$$\int_0^1 T_n^*(x')T_m(x')[x(1-x)]^{-1/2}\,dx = \frac{\delta_{mn}\pi}{2-\delta_{n0}}.$$

18.4.15 The expansion of a power of x in a Chebyshev series leads to the integral

$$I_{mn} = \int_{-1}^{1} x^m T_n(x)\frac{dx}{\sqrt{1-x^2}}.$$

(a) Show that this integral vanishes for $m < n$.
(b) Show that this integral vanishes for $m + n$ odd.

18.4.16 Evaluate the integral

$$I_{mn} = \int_{-1}^{1} x^m T_n(x)\frac{dx}{\sqrt{1-x^2}}$$

for $m \geq n$ and $m + n$ even by each of two methods:

(a) Replacing $T_n(x)$ by its Rodrigues representation.
(b) Using $x = \cos\theta$ to transform the integral to a form with θ as the variable.

$$\text{ANS.} \quad I_{mn} = \pi\frac{m!}{(m-n)!}\frac{(m-n-1)!!}{(m+n)!!}, \quad m \geq n,\ m+n \text{ even}.$$

18.4.17 Establish the following bounds, $-1 \leq x \leq 1$:

(a) $|U_n(x)| \leq n + 1$, (b) $\left|\frac{d}{dx}T_n(x)\right| \leq n^2$.

18.4.18 (a) Show that for $-1 \leq x \leq 1$, $|V_n(x)| \leq 1$.
(b) Show that $W_n(x)$ is unbounded in $-1 \leq x \leq 1$.

18.4.19 Verify the orthogonality-normalization integrals for

(a) $T_m(x)$, $T_n(x)$, (b) $V_m(x)$, $V_n(x)$,

(c) $U_m(x)$, $U_n(x)$, (d) $W_m(x)$, $W_n(x)$.

Hint. All these can be converted to trigonometric integrals.

18.4.20 Show whether

(a) $T_m(x)$ and $V_n(x)$ are or are not orthogonal over the interval $[-1, 1]$ with respect to the weighting factor $(1-x^2)^{-1/2}$.
(b) $U_m(x)$ and $W_n(x)$ are or are not orthogonal over the interval $[-1, 1]$ with respect to the weighting factor $(1-x^2)^{1/2}$.

18.4.21 Derive

(a) $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$,
(b) $T_{m+n}(x) + T_{m-n}(x) = 2T_m(x)T_n(x)$, from the "corresponding" cosine identities.

18.4.22 A number of equations relate the two types of Chebyshev polynomials. As examples show that

$$T_n(x) = U_n(x) - xU_{n-1}(x)$$

and

$$(1-x^2)U_n(x) = xT_{n+1}(x) - T_{n+2}(x).$$

18.4.23 Show that

$$\frac{dV_n(x)}{dx} = -n\frac{T_n(x)}{\sqrt{1-x^2}}$$

(a) using the trigonometric forms of V_n and T_n,
(b) using the Rodrigues representation.

18.4.24 Starting with $x = \cos\theta$ and $T_n(\cos\theta) = \cos n\theta$, expand

$$x^k = \left(\frac{e^{i\theta} + e^{-i\theta}}{2}\right)^k$$

and show that

$$x^k = \frac{1}{2^{k-1}}\left[T_k(x) + \binom{k}{1}T_{k-2}(x) + \binom{k}{2}T_{k-4} + \cdots\right],$$

the series in brackets terminating after the term containing T_1 or T_0.

18.4.25 Develop the following Chebyshev expansions (for $[-1, 1]$):

(a) $$(1-x^2)^{1/2} = \frac{2}{\pi}\left[1 - 2\sum_{s=1}^{\infty}(4s^2-1)^{-1}T_{2s}(x)\right],$$

(b) $$\left.\begin{array}{ll} +1, & 0 < x \le 1 \\ -1, & -1 \le x < 0 \end{array}\right\} = \frac{4}{\pi}\sum_{s=0}^{\infty}(-1)^s(2s+1)^{-1}T_{2s+1}(x).$$

18.4.26 (a) For the interval $[-1, 1]$ show that

$$|x| = \frac{1}{2} + \sum_{s=1}^{\infty}(-1)^{s+1}\frac{(2s-3)!!}{(2s+2)!!}(4s+1)P_{2s}(x)$$

$$= \frac{2}{\pi} + \frac{4}{\pi}\sum_{s=1}^{\infty}(-1)^{s+1}\frac{1}{4s^2-1}T_{2s}(x).$$

(b) Show that the ratio of the coefficient of $T_{2s}(x)$ to that of $P_{2s}(x)$ approaches $(\pi s)^{-1}$ as $s \to \infty$. This illustrates the relatively rapid convergence of the Chebyshev series.

Hint. With the Legendre recurrence relations, rewrite $x P_n(x)$ as a linear combination of derivatives. The trigonometric substitution $x = \cos\theta$, $T_n(x) = \cos n\theta$ is most helpful for the Chebyshev part.

18.4.27 Show that

$$\frac{\pi^2}{8} = 1 + 2\sum_{s=1}^{\infty}(4s^2-1)^{-2}.$$

Hint. Apply Parseval's identity (or the completeness relation) to the results of Exercise 18.4.26.

18.4.28 Show that

(a) $$\cos^{-1}x = \frac{\pi}{2} - \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}T_{2n+1}(x).$$

(b) $$\sin^{-1}x = \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}T_{2n+1}(x).$$

18.5 Hypergeometric Functions

In Chapter 7 the hypergeometric equation[8]

$$x(1-x)y''(x) + [c - (a+b+1)x]y'(x) - ab\,y(x) = 0 \tag{18.120}$$

[8]This is sometimes called Gauss' ODE. The solutions are then referred to as Gauss functions.

was introduced as a canonical form of a linear second-order ODE with regular singularities at $x = 0$, 1, and ∞. One solution, designated ${}_2F_1$, is

$$y(x) = {}_2F_1(a, b; c; x)$$
$$= 1 + \frac{a\,b}{c}\frac{x}{1!} + \frac{a(a+1)b(b+1)}{c(c+1)}\frac{x^2}{2!} + \cdots, \quad c \neq 0, -1, -2, -3, \ldots,$$

which is known as the **hypergeometric function** or **hypergeometric series**. For real a, b, and c (the only case considered here), the range of convergence for $c > a + b$ is $-1 \le x \le 1$, while for $a + b - 1 < c \le a + b$ the convergence range is $-1 \le x < 1$. For $c \le a + b - 1$ the hypergeometric series diverges.

The terms of the hypergeometric series are conveniently written in terms of the Pochhammer symbol, introduced at Eq. (1.72); we repeat the definition here:

$$(a)_n = a(a+1)(a+2)\cdots(a+n-1), \quad (a)_0 = 1.$$

Using this notation, the hypergeometric function becomes

$${}_2F_1(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{x^n}{n!}. \tag{18.121}$$

In this form the significance of the subscripts 2 and 1 becomes clear. The leading subscript 2 indicates that two Pochhammer symbols appear in the numerator and the trailing subscript 1 indicates one Pochhammer symbol in the denominator. The subscripts 2 and 1 are only useful if one intends to discuss analogs of the "standard" hypergeometric function that involve different numbers of Pochhammer symbols. We retain the subscripts because we will shortly identify **confluent hypergeometric functions** with forms similar to Eq. (18.121) but with only one Pochhammer symbol in the numerator, therefore of the form ${}_1F_1(a; c; z)$. Note also that the numerator and denominator parameters are set off by semicolons (actually making the subscripts unnecessary). We retain them to conform to the most widely used notations for these functions.

Looking further at Eq. (18.121), we note that the series will reduce to zero (for all x) if c is either zero or a negative integer (unless the denominator is fortuitously cancelled by a particular choice of a or b). On the other hand, if a or b equals 0 or a negative integer, the series terminates and the hypergeometric function becomes a polynomial. Many more or less elementary functions can be represented by the hypergeometric function.[9] For example,

$$\ln(1 + x) = x\,{}_2F_1(1, 1; 2; -x). \tag{18.122}$$

The hypergeometric equation as a second-order linear ODE has a second independent solution. The usual form is

$$y(x) = x^{1-c}\,{}_2F_1(a + 1 - c, b + 1 - c; 2 - c; x), \quad c \neq 2, 3, 4, \ldots. \tag{18.123}$$

If c is an integer either the two solutions coincide or (barring a rescue by integral a or integral b) one of the solutions will blow up (see Exercise 18.5.1). In such a case the second solution is expected to include a logarithmic term.

[9] With three parameters, a, b, and c, we can represent almost anything.

Alternate forms of the hypergeometric ODE include

$$(1-z^2)\frac{d^2}{dz^2}\left[\left(\frac{1-z}{2}\right)y\right]-\Big[(a+b+1)z-(a+b+1-2c)\Big]\frac{d}{dz}\left[\left(\frac{1-z}{2}\right)y\right]$$

$$-ab\left[\left(\frac{1-z}{2}\right)y\right]=0, \tag{18.124}$$

$$(1-z^2)\frac{d^2}{dz^2}y(z^2)-\left[(2a+2b+1)z+\frac{1-2c}{z}\right]\frac{d}{dz}y(z^2)-4ab\ y(z^2)=0. \tag{18.125}$$

Contiguous Function Relations

The parameters a, b, and c enter in the same way as the parameter n of Bessel, Legendre, and other special functions. As we found with these functions, we expect recurrence relations involving unit changes in the parameters a, b, and c. Hypergeometric functions that differ by ± 1 in a parameter are referred to as **contiguous functions**. Generalizing this term to include simultaneous unit changes in more than one parameter, we find 26 functions contiguous to ${}_2F_1(a, b; c; x)$. Taking them two at a time, we can develop the formidable total of 325 equations among the contiguous functions. Two typical examples are

$$(a-b)\Big\{c(a+b-1)+1-a^2-b^2+[(a-b)^2-1](1-x)\Big\}\ {}_2F_1(a,b;c;x)$$

$$=(c-a)(a-b+1)b\ {}_2F_1(a-1,b+1;c;x)$$

$$+(c-b)(a-b-1)a\ {}_2F_1(a+1,b-1;c;x), \tag{18.126}$$

$$[2a-c+(b-a)x]\,{}_2F_1(a,b;c;x)=a(1-x)\,{}_2F_1(a+1,b;c;x)$$

$$-(c-a)\,{}_2F_1(a-1,b;c;x). \tag{18.127}$$

Many more contiguous relations can be found in AMS-55 or in Olver *et al*. (Additional Readings).

Hypergeometric Representations

A number of the special functions introduced in this book can be expressed in terms of hypergeometric functions. The identification can usually be made by noting that these functions are solutions of ODEs that are special cases of the hypergeometric ODE. It is also necessary to determine the factors needed to express the functions at the agreed-upon scale. We cite several examples.

1. The ultraspherical functions $C_n^{(\alpha)}(x)$ satisfy the ODE given as Eq. (18.99), and since that equation is a special case of the hypergeometric equation, Eq. (18.120), we see that ultraspherical functions (and Legendre and Chebyshev functions) may be expressed as

hypergeometric functions. For the ultraspherical function we obtain

$$C_n^{(\alpha)}(x) = \frac{(n+2\alpha)!}{2^\alpha n!\,\Gamma(\alpha+1)}\,{}_2F_1\left(-n, n+2\alpha+1; 1+\alpha; \frac{1-x}{2}\right), \tag{18.128}$$

with the factor preceding the ${}_2F_1$ function determined by requiring $C_n^{(\alpha)}$ to have the proper scale.

2. For Legendre and associated Legendre functions we find

$$P_n(x) = {}_2F_1\left(-n, n+1; 1; \frac{1-x}{2}\right), \tag{18.129}$$

$$P_n^m(x) = \frac{(n+m)!}{(n-m)!}\frac{(1-x^2)^{m/2}}{2^m m!}\,{}_2F_1\left(m-n, m+n+1; m+1; \frac{1-x}{2}\right). \tag{18.130}$$

Alternate forms for the Legendre functions are

$$\begin{aligned} P_{2n}(x) &= (-1)^n \frac{(2n)!}{2^{2n} n!\, n!}\,{}_2F_1\left(-n, n+\frac{1}{2}; \frac{1}{2}; x^2\right) \\ &= (-1)^n \frac{(2n-1)!!}{(2n)!!}\,{}_2F_1\left(-n, n+\frac{1}{2}; \frac{1}{2}; x^2\right), \end{aligned} \tag{18.131}$$

$$\begin{aligned} P_{2n+1}(x) &= (-1)^n \frac{(2n+1)!}{2^{2n} n!\, n!}\,x\,{}_2F_1\left(-n, n+\frac{3}{2}; \frac{3}{2}; x^2\right) \\ &= (-1)^n \frac{(2n+1)!!}{(2n)!!}\,x\,{}_2F_1\left(-n, n+\frac{3}{2}; \frac{3}{2}; x^2\right). \end{aligned} \tag{18.132}$$

3. The Chebyshev functions have representations

$$T_n(x) = {}_2F_1\left(-n, n; \frac{1}{2}; \frac{1-x}{2}\right), \tag{18.133}$$

$$U_n(x) = (n+1)\,{}_2F_1\left(-n, n+2; \frac{3}{2}; \frac{1-x}{2}\right), \tag{18.134}$$

$$V_n(x) = n\sqrt{1-x^2}\,{}_2F_1\left(-n+1, n+1; \frac{3}{2}; \frac{1-x}{2}\right). \tag{18.135}$$

The leading factors are determined by direct comparison of complete power series, comparison of coefficients of particular powers of the variable, or evaluation at $x = 0$ or 1.

The hypergeometric series may be used to define functions with nonintegral indices. The physical applications are minimal.

Exercises

18.5.1 (a) For c, an integer, and a and b nonintegral, show that

$$_2F_1(a,b;c;x) \quad \text{and} \quad x^{1-c}\,{}_2F_1(a+1-c,b+1-c;2-c;x)$$

yield only one solution to the hypergeometric equation.

(b) What happens if a is an integer, say, $a=-1$, and $c=-2$?

18.5.2 Find the Legendre, Chebyshev I, and Chebyshev II recurrence relations corresponding to the hypergeometric contiguous function relation given as Eq. (18.126).

18.5.3 Transform the following polynomials into hypergeometric functions of argument x^2:

(a) $T_{2n}(x)$;

(b) $x^{-1}T_{2n+1}(x)$;

(c) $U_{2n}(x)$;

(d) $x^{-1}U_{2n+1}(x)$.

ANS. (a) $T_{2n}(x) = (-1)^n\,{}_2F_1(-n,n;\frac{1}{2};x^2)$.
(b) $x^{-1}T_{2n+1}(x) = (-1)^n(2n+1)\,{}_2F_1\big(-n,n+1;\frac{3}{2};x^2\big)$.
(c) $U_{2n}(x) = (-1)^n\,{}_2F_1\big(-n,n+1;\frac{1}{2};x^2\big)$.
(d) $x^{-1}U_{2n+1}(x) = (-1)^n(2n+2)\,{}_2F_1\big(-n,n+2;\frac{3}{2};x^2\big)$.

18.5.4 Derive or verify the leading factor in the hypergeometric representations of the Chebyshev functions.

18.5.5 Verify that the Legendre function of the second kind, $Q_\nu(z)$, is given by

$$Q_\nu(z) = \frac{\pi^{1/2}\nu!}{\Gamma(\nu+\frac{3}{2})(2z)^{\nu+1}}\,{}_2F_1\left(\frac{\nu}{2}+\frac{1}{2},\frac{\nu}{2}+1;\frac{\nu}{2}+\frac{3}{2};z^{-2}\right),$$

where $|z|>1$, $|\arg z|<\pi$, and $\nu \neq -1,-2,-3,\cdots$.

18.5.6 The incomplete beta function was defined in Eq. (13.78) as

$$B_x(p,q) = \int_0^x t^{p-1}(1-t)^{q-1}\,dt.$$

Show that

$$B_x(p,q) = p^{-1}x^p\,{}_2F_1(p,1-q;p+1;x).$$

18.5.7 Verify the integral representation

$$_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)}\int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\,dt.$$

What restrictions must be placed on the parameters b and c?

Note. Although the power series used to establish this integral representation is only valid for $|z| < 1$, the representation is valid for general z, as can be established by analytic continuation. For nonintegral a the real axis in the z-plane from 1 to ∞ is a cut line.

Hint. The integral is suspiciously like a beta function and can be expanded into a series of beta functions.

ANS. $c > b > 0$.

18.5.8 Prove that

$$ {}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}, \quad c \neq 0, -1, -2, \ldots, \quad c > a + b. $$

Hint. Here is a chance to use the integral representation in Exercise 18.5.7.

18.5.9 Prove that

$$ {}_2F_1(a, b; c; x) = (1-x)^{-a}\, {}_2F_1\left(a, c-b; c; \frac{-x}{1-x}\right). $$

Hint. Try an integral representation.

Note. This relation is useful in developing a Rodrigues representation of $T_n(x)$ (see Exercise 18.5.10).

18.5.10 Derive the Rodrigues representation of $T_n(x)$,

$$ T_n(x) = \frac{(-1)^n \pi^{1/2}(1-x^2)^{1/2}}{2^n(n-\frac{1}{2})!} \frac{d^n}{dx^n}\left[(1-x^2)^{n-1/2}\right]. $$

Hint. One possibility is to use the hypergeometric function relation

$$ {}_2F_1(a, b; c; z) = (1-z)^{-a}\, {}_2F_1\left(a, c-b; c; \frac{-z}{1-z}\right), $$

with $z = (1-x)/2$. An alternate approach is to develop a first-order differential equation for $y = (1-x^2)^{n-1/2}$. Repeated differentiation of this equation leads to the Chebyshev equation.

18.5.11 Show that the summation in Eq. (18.43),

$$ \sum_{\nu=0}^{\min(m,n)} \frac{(-m)_\nu(-n)_\nu}{\nu!\left(\frac{1-m-n}{2}\right)_\nu}\left(\frac{a^2}{2(a^2-1)}\right)^\nu, $$

can be written as a hypergeometric function.

18.5.12 Verify that

$$ {}_2F_1(-n, b; c; 1) = \frac{(c-b)_n}{(c)_n}. $$

Hint. Here is a chance to use the contiguous function relation Eq. (18.127) and mathematical induction (Section 1.4). Alternatively, use the integral representation and the beta function.

18.6 Confluent Hypergeometric Functions

The confluent hypergeometric equation,[10]

$$xy''(x) + (c - x)y'(x) - ay(x) = 0, \tag{18.136}$$

has a regular singularity at $x = 0$ and an irregular one at $x = \infty$. It is obtained from the hypergeometric equation of Section 18.5 in the limit that one of the singularities at finite x is merged with that at infinity, causing that singularity to become irregular. One solution of the confluent hypergeometric equation is

$$\begin{aligned} y(x) &= {}_1F_1(a; c; x) = M(a, c, x) \\ &= 1 + \frac{a}{c}\frac{x}{1!} + \frac{a(a+1)}{c(c+1)}\frac{x^2}{2!} + \cdots, \quad c \neq 0, -1, -2, \cdots. \end{aligned} \tag{18.137}$$

The notation $M(a, c, x)$ (with commas, not semicolons) has become standard for this solution. It is convergent for all finite x (or complex z). In terms of the Pochhammer symbols, we have

$$M(a, c, x) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n}\frac{x^n}{n!}. \tag{18.138}$$

Clearly, $M(a, c, x)$ becomes a polynomial if the parameter a is 0 or a negative integer. Numerous more or less elementary functions may be represented by the confluent hypergeometric function. Examples are the error function and the incomplete gamma function:

$$\operatorname{erf}(x) = \frac{2}{\pi^{1/2}} \int_0^x e^{-t^2} dt = \frac{2}{\pi^{1/2}}\, x\, M\left(\frac{1}{2}, \frac{3}{2}, -x^2\right), \tag{18.139}$$

$$\gamma(a, x) = \int_0^x e^{-t} t^{a-1} dt = a^{-1} x^a M(a, a+1, -x), \quad \Re e(a) > 0. \tag{18.140}$$

A second solution of Eq. (18.136) is given by

$$y(x) = x^{1-c} M(a+1-c, 2-c, x), \quad c \neq 2, 3, 4, \cdots. \tag{18.141}$$

Clearly, this coincides with the first solution for $c = 1$.

The standard form of the second solution of Eq. (18.136) is a linear combination of Eqs. (18.137) and (18.141):

$$U(a, c, x) = \frac{\pi}{\sin \pi c}\left[\frac{M(a, c, x)}{\Gamma(a - c + 1)\Gamma(c)} - \frac{x^{1-c} M(a + 1 - c, 2 - c, x)}{\Gamma(a)\Gamma(-c)}\right]. \tag{18.142}$$

Note the resemblance to our definition of the Neumann function, Eq. (14.57). As with the Neumann function, this definition of $U(a, c, x)$ becomes indeterminate for certain parameter values, namely when c is an integer.

[10]This is often called **Kummer's equation**. The solutions, then, are **Kummer functions**.

An alternate form of the confluent hypergeometric equation is obtained by changing the independent variable from x to x^2:

$$\frac{d^2}{dx^2}y(x^2)+\left[\frac{2c-1}{x}-2x\right]\frac{d}{dx}y(x^2)-4ay(x^2)=0. \tag{18.143}$$

As with the hypergeometric functions, contiguous functions exist in which the parameters a and c are changed by ± 1. Including the cases of simultaneous changes in the two parameters, we have eight possibilities. Taking the original function and pairs of the contiguous functions, we can develop a total of 28 equations. The recurrence relations for Bessel, Hermite, and Laguerre functions are special cases of these equations.

Integral Representations

It is frequently convenient to have the confluent hypergeometric functions in integral form. We find (Exercise 18.6.10)

$$M(a,c,x)=\frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\int_0^1 e^{xt}t^{a-1}(1-t)^{c-a-1}dt, \quad c>a>0, \tag{18.144}$$

$$U(a,c,x)=\frac{1}{\Gamma(a)}\int_0^\infty e^{-xt}t^{a-1}(1+t)^{c-a-1}dt, \quad \Re e(x)>0, a>0. \tag{18.145}$$

Three important techniques for deriving or verifying integral representations are as follows:

1. Transformation of generating function expansions and Rodrigues representations: The Bessel and Legendre functions provide examples of this approach.
2. Direct integration to yield a series: This direct technique is useful for a Bessel function representation (Exercise 14.1.17) and a hypergeometric integral (Exercise 18.5.7).
3. (a) Verification that the integral representation satisfies the ODE. (b) Exclusion of the other solution. (c) Verification of normalization. This is the method used in Section 14.6 to establish an integral representation of the modified Bessel function $K_\nu(z)$. It will work here to establish Eqs. (18.144) and (18.145).

Confluent Hypergeometric Representations

Special functions that can be represented in terms of confluent hypergeometric functions include the following:

1. **Bessel functions**:

$$J_\nu(x)=\frac{e^{-ix}}{\Gamma(\nu+1)}\left(\frac{x}{2}\right)^\nu M\left(\nu+\frac{1}{2},2\nu+1,2ix\right), \tag{18.146}$$

whereas for the modified Bessel functions of the first kind,

$$I_\nu(x) = \frac{e^{-x}}{\Gamma(\nu+1)}\left(\frac{x}{2}\right)^\nu M\left(\nu+\frac{1}{2}, 2\nu+1, 2x\right). \tag{18.147}$$

2. **Hermite functions**:

$$H_{2n}(x) = (-1)^n \frac{(2n)!}{n!} M\left(-n, \frac{1}{2}, x^2\right). \tag{18.148}$$

$$H_{2n+1}(x) = (-1)^n \frac{2(2n+1)!}{n!} x M\left(-n, \frac{3}{2}, x^2\right), \tag{18.149}$$

using Eq. (13.150).

3. **Laguerre functions**:

$$L_n(x) = M(-n, 1, x). \tag{18.150}$$

The constant is fixed as unity by noting Eq. (18.54) for $x = 0$. For the associated Laguerre functions,

$$L_n^m(x) = (-1)^m \frac{d^m}{dx^m} L_{n+m}(x) = \frac{(n+m)!}{n!\,m!} M(-n, m+1, x). \tag{18.151}$$

Alternate verification is obtained by comparing Eq. (18.151) with the power-series solution, Eq. (18.59). Note that in the hypergeometric form, as distinct from a Rodrigues representation, the indices n and m need not be integers, but if they are not integers, $L_n^m(x)$ will not be a polynomial.

Further Observations

There are certain advantages in expressing our special functions in terms of hypergeometric and confluent hypergeometric functions. If the general behavior of the latter functions is known, the behavior of the special functions we have investigated follows as a series of special cases. This may be useful in determining asymptotic behavior or evaluating normalization integrals. The asymptotic behavior of $M(a,c,x)$ and $U(a,c,x)$ may be conveniently obtained from integral representations of these functions, Eqs. (18.144) and (18.145). The further advantage is that the relations between the special functions are clarified. For instance, an examination of Eqs. (18.148), (18.149), and (18.151) suggests that the Laguerre and Hermite functions are related.

The confluent hypergeometric equation, Eq. (18.136), is clearly not self-adjoint. For this and other reasons it is convenient to define

$$M_{k\mu}(x) = e^{-x/2} x^{\mu+1/2} M(\mu - k + \tfrac{1}{2}, 2\mu+1, x). \tag{18.152}$$

This new function, $M_{k\mu}(x)$, is called a Whittaker function; it satisfies the self-adjoint equation

$$M''_{k\mu}(x) + \left(-\frac{1}{4} + \frac{k}{x} + \frac{\frac{1}{4}-\mu^2}{x^2}\right) M_{k\mu}(x) = 0. \tag{18.153}$$

The corresponding second solution is

$$W_{k\mu}(x) = e^{-x/2} x^{\mu+1/2} U(\mu - k + \tfrac{1}{2}, 2\mu + 1, x). \tag{18.154}$$

Exercises

18.6.1 Verify the confluent hypergeometric representation of the error function

$$\operatorname{erf}(x) = \frac{2x}{\pi^{1/2}} M\left(\frac{1}{2}, \frac{3}{2}, -x^2\right).$$

18.6.2 Show that the Fresnel integrals $C(x)$ and $s(x)$ of Exercise 12.6.1 may be expressed in terms of the confluent hypergeometric function as

$$C(x) + is(x) = xM\left(\frac{1}{2}, \frac{3}{2}, \frac{i\pi x^2}{2}\right).$$

18.6.3 By direct differentiation and substitution verify that

$$y = ax^{-a} \int_0^x e^{-t} t^{a-1}\, dt = ax^{-a} \gamma(a, x)$$

satisfies

$$xy'' + (a + 1 + x)y' + ay = 0.$$

18.6.4 Show that the modified Bessel function of the second kind, $K_\nu(x)$, is given by

$$K_\nu(x) = \pi^{1/2} e^{-x} (2x)^\nu U(\nu + \tfrac{1}{2}, 2\nu + 1, 2x).$$

18.6.5 Show that the cosine and sine integrals of Section 13.6 may be expressed in terms of confluent hypergeometric functions as

$$\operatorname{Ci}(x) + i \operatorname{si}(x) = -e^{ix} U(1, 1, -ix).$$

This relation is useful in numerical computation of $\operatorname{Ci}(x)$ and $\operatorname{si}(x)$ for large values of x.

18.6.6 Verify the confluent hypergeometric form of the Hermite polynomial $H_{2n+1}(x)$, Eq. (18.149), by showing that

(a) $H_{2n+1}(x)/x$ satisfies the confluent hypergeometric equation with $a = -n$, $c = 3/2$ and argument x^2,

(b) $\displaystyle\lim_{x\to 0} \frac{H_{2n+1}(x)}{x} = (-1)^n \frac{2(2n+1)!}{n!}.$

18.6.7 Show that the contiguous confluent hypergeometric function equation

$$(c - a)M(a - 1, c, x) + (2a - c + x)M(a, c, x) - aM(a + 1, c, x) = 0$$

leads to the associated Laguerre function recurrence relation, Eq. (18.66).

18.6.8 Verify the Kummer transformations:

(a) $M(a,c,x) = e^x M(c-a,c,-x)$,
(b) $U(a,c,x) = x^{1-c} U(a-c+1, 2-c, x)$.

18.6.9 Prove that

(a) $\dfrac{d^n}{dx^n} M(a,c,x) = \dfrac{(a)_n}{(b)_n} M(a+n, b+n, x)$,
(b) $\dfrac{d^n}{dx^n} U(a,c,x) = (-1)^n (a)_n U(a+n, c+n, x)$.

18.6.10 Verify the following integral representations:

(a) $M(a,c,x) = \dfrac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)} \displaystyle\int_0^1 e^{xt} t^{a-1} (1-t)^{c-a-1} dt, c > a > 0$,
(b) $U(a,c,x) = \dfrac{1}{\Gamma(a)} \displaystyle\int_0^\infty e^{-xt} t^{a-1} (1+t)^{c-a-1} dt, \quad \Re e(x) > 0, a > 0.$

Under what conditions can you accept $\Re e(x) = 0$ in part (b)?

18.6.11 From the integral representation of $M(a,c,x)$, Exercise 18.6.10(a), show that

$$M(a,c,x) = e^x M(c-a,c,-x).$$

Hint. Replace the variable of integration t by $1-s$ to release a factor e^x from the integral.

18.6.12 From the integral representation of $U(a,c,x)$ in Exercise 18.6.10(b), show that the exponential integral is given by

$$E_1(x) = e^{-x} U(1,1,x).$$

Hint. Replace the variable of integration t in $E_1(x)$ by $x(1+s)$.

18.6.13 From the integral representations of $M(a,c,x)$ and $U(a,c,x)$ in Exercise 18.6.10, develop asymptotic expansions of

(a) $M(a,c,x)$, (b) $U(a,c,x)$.

Hint. You can use the technique that was employed with $K_\nu(z)$ in Section 14.6.

ANS.
(a) $\dfrac{\Gamma(c)}{\Gamma(a)} \dfrac{e^x}{x^{c-a}} \left\{ 1 + \dfrac{(1-a)(c-a)}{1!\,x} + \dfrac{(1-a)(2-a)(c-a)(c-a+1)}{2!\,x^2} + \cdots \right\}$,

(b) $\dfrac{1}{x^a} \left\{ 1 + \dfrac{a(1+a-c)}{1!\,(-x)} + \dfrac{a(a+1)(1+a-c)(2+a-c)}{2!\,(-x)^2} + \cdots \right\}$.

18.6.14 Show that the Wronskian of the two confluent hypergeometric functions $M(a,c,x)$ and $U(a,c,x)$ is given by

$$MU' - M'U = -\frac{(c-1)!}{(a-1)!}\frac{e^x}{x^c}.$$

What happens if a is 0 or a negative integer?

18.6.15 The Coulomb wave equation (radial part of the Schrödinger equation with Coulomb potential) is

$$\frac{d^2y}{dr^2} + \left[1 - \frac{2\eta}{r} - \frac{L(L+1)}{r^2}\right] y = 0.$$

Show that a regular solution $y = F_L(\eta, r)$ is given by

$$F_L(\eta, r) = C_L(\eta) r^{L+1} e^{-ir} M(L+1-i\eta, 2L+2, 2ir).$$

18.6.16 (a) Show that the radial part of the hydrogen-atom wave function, Eq. (18.81), may be written as

$$e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r) =$$

$$\frac{(n+L)!}{(n-L-1)!(2L+1)!} e^{-\alpha r/2}(\alpha r)^L M(L+1-n, 2L+2, \alpha r).$$

(b) It was assumed previously that the total (kinetic + potential) energy E of the electron was negative. Rewrite the (unnormalized) radial wave function for an unbound hydrogenic electron, $E > 0$.

ANS. $e^{i\alpha r/2}(\alpha r)^L M(L+1-in, 2L+2, -i\alpha r)$, outgoing wave. This representation provides a powerful alternative technique for the calculation of photoionization and recombination coefficients.

18.6.17 Evaluate

$$\text{(a)} \int_0^\infty [M_{k\mu}(x)]^2 dx, \quad \text{(b)} \int_0^\infty [M_{k\mu}(x)]^2 \frac{dx}{x}, \quad \text{(c)} \int_0^\infty [M_{k\mu}(x)]^2 \frac{dx}{x^{1-a}},$$

where $2\mu = 0, 1, 2, \ldots, \quad k - \mu - \frac{1}{2} = 0, 1, 2, \ldots, \quad a > -2\mu - 1.$

ANS. (a) $2k(2\mu)!$, (b) $(2\mu)!$, (c) $(2k)^a(2\mu)!$.

18.7 DILOGARITHM

The **dilogarithm**, defined as

$$\mathrm{Li}_2(z) = -\int_0^z \frac{\ln(1-t)}{t}dt \tag{18.155}$$

and its analytic continuation beyond the range of convergence of the above integral, arises in the evaluation of matrix elements in few-body problems of atomic physics and in various perturbation-theoretic contributions to quantum electrodynamics. Because of a historic lack of familiarity with this special function among physicists, many places of its occurrence have only been recognized in recent years.

Expansion and Analytic Properties

Expanding the logarithm in Eq. (18.155), using the series in Eq. (1.97), we directly obtain the series expansion

$$\mathrm{Li}_2(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^2}. \tag{18.156}$$

Note that we have inserted the logarithm without an additional multiple of $2\pi i$, thereby obtaining the branch of Li_2 that is nonsingular at $z = 0$.

Further applications of the operator that converts $-\ln(1-z)$ into $\mathrm{Li}_2(z)$ produce **polylogarithms**, which also occur in physics, albeit less frequently:

$$\mathrm{Li}_p(z) = \int_0^z \mathrm{Li}_{p-1}(t)\frac{dt}{t} = \sum_{n=1}^{\infty} \frac{z^n}{n^p} \quad p = 3,\ 4, \ldots. \tag{18.157}$$

However, in this text we limit consideration to the first member of this sequence, Li_2.

The series expansion of Li_2, Eq. (18.156), has circle of convergence $|z| = 1$, with convergence for all z on this circle. The singularity limiting the radius of convergence is not apparent from the form of the expansion, but, looking at Eq. (18.155), we identify it as a branch point located at $z = 1$. It is customary to draw a branch cut from $z = 1$ to $z = \infty$ along, and just below the positive real axis, and to define the principal value of Li_2 as that which corresponds to Eq. (18.156) and its analytic continuation.

From the form of Eq. (18.156), it is apparent that for real z in the interval $-1 \le z \le +1$, $\mathrm{Li}_2(z)$ will also be real. For $z > 1$, we see from Eq. (18.155) that for part of the range of integration, the factor $\ln(1-t)$ will necessarily be complex, with the result that $\mathrm{Li}_2(z)$ will no longer be real, even for real z. However, there is no similar problem for negative real z, as the principal value of $\ln(1-t)$ remains real for all negative real values of t.

Analyzing further the behavior of the integral in Eq. (18.155), we note that if we reach a point z by carrying out the integral, along a path (in t) that goes first from $t = 0$ to just above the branch point at $t = 1$, and then in a straight line to z, we will for the last segment of the path alter the argument of $1-t$ by some amount θ in the clockwise direction, thereby adding an amount $-i\theta$ to the numerator of the integrand. See Fig. 18.8.

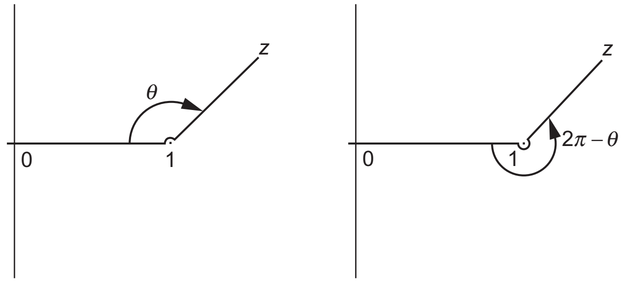

FIGURE 18.8 Contours for integral representation of dilogarithm.

This addition to the numerator means that the evaluation of $\mathrm{Li}_2(z)$ will have the form

$$\mathrm{Li}_2(z) = -\int_0^1 \frac{\ln(1-t)}{t}\,dt - \int_1^z \frac{\ln(|1-t|)}{t}\,dt + i\theta \int_1^z \frac{dt}{t}$$

$$= \mathrm{Li}_2(1) - \int_1^z \frac{\ln(|1-t|)}{t}\,dt + i\theta \ln z \quad \text{(path above } z = 1\text{)}. \tag{18.158}$$

If we repeat the above analysis to reach the same point z by a path (in t) that passes around $z = 1$ below the real axis, the argument of $1 - t$ will be changed by an amount $2\pi - \theta$ in the counterclockwise direction, and

$$\mathrm{Li}_2(z) = \mathrm{Li}_2(1) - \int_1^z \frac{\ln(|1-t|)}{t}\,dt - i(2\pi - \theta)\ln z \quad \text{(path below } z = 1\text{)}. \tag{18.159}$$

Comparing Eqs. (18.158) and (18.159), we see that the values of $\mathrm{Li}_2(z)$, for the same z, but on these two different branches the values, will differ by an amount $2\pi i \ln z$. If z is complex, the difference will affect both the real and imaginary parts of $\mathrm{Li}_2(z)$, in ways more complicated than either changing the phase or adding a multiple of π to the imaginary part. When working with the dilogarithm, it is therefore essential to make a careful determination of the branch on which it is to be evaluated. In fact, whenever possible formulas involving the dilogarithm and (because of the context) known to be real-valued should be manipulated (using formulas such as those in the next subsection) to cause each dilogarithm in the formula to be for a value of z that is real and with $z < 1$.

Properties and Special Values

From Eq. (18.156), we see that $\mathrm{Li}_2(0) = 0$. Setting $z = 1$, we note that we get the series for $\zeta(2)$, so $\mathrm{Li}_2(1) = \zeta(2) = \pi^2/6$. We also have $\mathrm{Li}_2(-1) = -\eta(2)$, where $\eta(2)$ is the Dirichlet series in Eq. (12.62), so $\mathrm{Li}_2(-1) = -\pi^2/12$.

The dilogarithm has a derivative that follows directly from Eq. (18.155),

$$\frac{d\,\mathrm{Li}_2(z)}{dz} = -\frac{\ln(1-z)}{z}, \tag{18.160}$$

and possesses several functional relations enabling an easy analytic continuation beyond the convergence range of Eq. (18.156). Some of these are the following:

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z) \tag{18.161}$$

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(z^{-1}) = -\frac{\pi^2}{6} - \frac{1}{2}\ln^2(-z) \tag{18.162}$$

$$\mathrm{Li}_2(z) + \mathrm{Li}_2\left(\frac{z}{z-1}\right) = -\frac{1}{2}\ln^2(1-z). \tag{18.163}$$

These relationships are most easily established by showing that the derivatives of both sides of the equations are equal and that the values of the two sides correspond for some convenient value of z. These functional relations enable the determination of $\mathrm{Li}_2(z)$ for all real z from values on the real line in the range $|z| \le \frac{1}{2}$, for which the series in Eq. (18.155) converges rapidly.

From the functional relations it is possible to identify a few more specific values of z for which the principal value of $\mathrm{Li}_2(z)$ can be expressed in terms of elementary functions. For example, $\mathrm{Li}_2(1/2) = -\frac{1}{2}\ln^2(2) + \pi^2/12$. But for most z, closed expressions are not available.

Example 18.7.1 Check Usefulness of Formula

The integral

$$I = \frac{1}{8\pi^2}\iint d^3r_1 d^3r_2\, \frac{e^{-\alpha r_1 - \beta r_2 - \gamma r_{12}}}{r_1^2 r_2^2 r_{12}}$$

arises in computations of the electronic structure of the He atom. Here $\mathbf{r}_i$ are the positions of two electrons relative to the nucleus (which is at the origin of our coordinate system), the integration is over the full three-dimensional spaces of $\mathbf{r}_1$ and $\mathbf{r}_2$, $r_i = |\mathbf{r}_i|$, and $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$.

This integral is found to have the value

$$I = \frac{1}{\gamma}\left[\frac{\pi^2}{6} + \mathrm{Li}_2\left(\frac{\gamma-\beta}{\alpha+\gamma}\right) + \mathrm{Li}_2\left(\frac{\gamma-\alpha}{\beta+\gamma}\right) + \frac{1}{2}\ln^2\left(\frac{\alpha+\gamma}{\beta+\gamma}\right)\right].$$

We now ask: Are its individual terms real?

We note from the definition of I that it will be convergent only if $\alpha+\beta$, $\alpha+\gamma$, and $\beta+\gamma$ are all positive. If that is not the case, in the portion of the space in which some particle is far from the other two, the overall exponential will increase without limit. Looking now at the formula for the integral, we see immediately that the $\ln^2$ term will be real, as its argument is the quotient of two positive numbers. The first Li_2 term can be written

$$\mathrm{Li}_2\left(\frac{\gamma-\beta}{\alpha+\gamma}\right)=\mathrm{Li}_2\left(1-\frac{\alpha+\beta}{\alpha+\gamma}\right),$$

showing that the argument of Li_2 is real and less than $+1$, meaning that this Li_2 will evaluate to a real result. Similar observations apply to the second instance of Li_2. We conclude that our formula is in a proper form for unambiguous computation using principal values of its multivalued functions. ■

Exercises

18.7.1 Prove that the expansion of $\mathrm{Li}_2(z)$, Eq. (18.156), converges everywhere on the circle $|z|=1$.

18.7.2 Use the functional relations, Eqs. (18.161) to (18.163), to find the principal value of $\mathrm{Li}_2(1/2)$.

18.7.3 Find all the multiple values of $\mathrm{Li}_2(1/2)$.

18.7.4 Explain why Eq. (18.161) gives the expected result for $z=0$ when on the principal branch of the dilogarithm.

18.7.5 Show that

$$\mathrm{Li}_2\left(\frac{1+z^{-1}}{2}\right)=-\mathrm{Li}_2\left(\frac{1+z}{1-z}\right)-\frac{1}{2}\ln^2\left(\frac{1-z^{-1}}{2}\right).$$

18.7.6 The following integral arises in the computation of the electronic energy of the Li atom using a correlated wave function (one that explicitly includes the electron-electron distances as well as the distances of electrons from the nucleus):

$$I=\iiint d^3r_1d^3r_2d^3r_3\,\frac{e^{-\alpha_1r_1-\alpha_2r_2-\alpha_3r_3}}{r_1r_2r_3r_{12}r_{13}r_{23}},$$

where $r_i=|\mathbf{r}_i|$, $r_{ij}=|\mathbf{r}_i-\mathbf{r}_j|$, and the integrations are over the entire three-dimensional space of each $\mathbf{r}_i$. For convergence of I, we require all $\alpha_j>0$, but there are no restrictions on their relative magnitudes.

In terms of the auxiliary quantities

$$\zeta_1=\frac{\alpha_1}{\alpha_2+\alpha_3},\qquad \zeta_2=\frac{\alpha_2}{\alpha_1+\alpha_3},\qquad \zeta_3=\frac{\alpha_3}{\alpha_1+\alpha_2},$$

this integral has the value

$$I=\frac{32\pi^3}{\alpha_1\alpha_2\alpha_3}\left(-\frac{\pi^2}{2}+\sum_{j=1}^{3}\left[\mathrm{Li}_2(\zeta_j)-\mathrm{Li}_2(-\zeta_j)+\ln\zeta_j\ln\left(\frac{1-\zeta_j}{1+\zeta_j}\right)\right]\right)$$

Rearrange I to a form (first found by Remiddi[11]), in which all terms in the final expression are guaranteed to evaluate to real quantities and can be evaluated as principal values.

18.8 ELLIPTIC INTEGRALS

Elliptic integrals occasionally arise in physical problems and therefore it is worthwhile to summarize their definitions and properties. Before the advent of computers, it was also important for physicists and engineers to be familiar with methods for hand computation of elliptic integrals, but that need has diminished with time and expansion methods for these functions will not be emphasized here. We do, however, illustrate problems in which elliptic integrals arise; the following example is a case in point.

Example 18.8.1 PERIOD OF A SIMPLE PENDULUM

For small-amplitude oscillations, a pendulum (Fig. 18.9) has simple harmonic motion with a period $T = 2\pi(l/g)^{1/2}$. But for a maximum amplitude θ_M large enough that $\sin\theta_M$ cannot be approximated by θ_M, a direct application of Newton's second law of motion and solution of the resulting ODE becomes difficult. In that situation a good way to proceed is to write the equation for conservation of energy. Setting the zero of potential energy at the point from which the pendulum is suspended, the potential energy of a pendulum of mass m and length l at angle θ is $-mgl\cos\theta$, and its total energy (the potential energy at angle θ_M) is $-mgl\cos\theta_M$. The pendulum has kinetic energy $ml^2(d\theta/dt)^2/2$, so energy conservation requires

$$\frac{1}{2}ml^2\left(\frac{d\theta}{dt}\right)^2 - mgl\cos\theta = -mgl\cos\theta_M. \tag{18.164}$$

Solving for $d\theta/dt$ we obtain

$$\frac{d\theta}{dt} = \pm\left(\frac{2g}{l}\right)^{1/2}(\cos\theta - \cos\theta_M)^{1/2}, \tag{18.165}$$

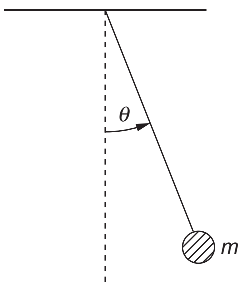

FIGURE 18.9 Simple pendulum.

[11] E. Remiddi, Analytic value of the atomic three-electron correlation integral with Slater wave functions. *Phys. Rev. A* **44**: 5492 (1991).

with the mass m canceling out. At $t = 0$ we choose as initial conditions $\theta = 0$ and $d\theta/dt > 0$. An integration from $\theta = 0$ to $\theta = \theta_M$ yields

$$\int_0^{\theta_M} (\cos\theta - \cos\theta_M)^{-1/2} d\theta = \left(\frac{2g}{l}\right)^{1/2} \int_0^t dt = \left(\frac{2g}{l}\right)^{1/2} t. \tag{18.166}$$

This is $\frac{1}{4}$ of a cycle, and therefore the time t is $\frac{1}{4}$ of the period T. We note that $\theta \le \theta_M$, and with a bit of clairvoyance we try the half-angle substitution

$$\sin\left(\frac{\theta}{2}\right) = \sin\left(\frac{\theta_M}{2}\right) \sin\varphi. \tag{18.167}$$

With this, Eq. (18.166) becomes

$$T = 4\left(\frac{l}{g}\right)^{1/2} \int_0^{\pi/2} \left(1 - \sin^2\left(\frac{\theta_M}{2}\right) \sin^2\varphi\right)^{-1/2} d\varphi. \tag{18.168}$$

The integral in Eq. (18.168) does not reduce to an elementary function; in fact, it is an **elliptic integral** of a standard type. Further examples of elliptic integrals in physical problems can be found in the exercises. ■

Definitions

The **elliptic integral of the first kind** is defined as

$$F(\varphi\backslash\alpha) = \int_0^{\varphi} (1 - \sin^2\alpha \sin^2\theta)^{-1/2} d\theta, \tag{18.169}$$

or

$$F(x|m) = \int_0^x \left[(1 - t^2)(1 - mt^2)\right]^{-1/2} dt, \quad 0 \le m < 1. \tag{18.170}$$

This is the notation of AMS-55 (Additional Readings). Note the use of the separators \ and | to identify the specific functional forms. When the upper limit in these integrals is set to $\varphi = \pi/2$ or $x = 1$, we have the **complete elliptic integral of the first kind**,

$$\begin{aligned} K(m) &= \int_0^{\pi/2} (1 - m\sin^2\theta)^{-1/2} d\theta \\ &= \int_0^1 \left[(1 - t^2)(1 - mt^2)\right]^{-1/2} dt, \end{aligned} \tag{18.171}$$

with $m = \sin^2\alpha$, $0 \le m < 1$.

The **elliptic integral of the second kind** is defined by

$$E(\varphi\backslash\alpha) = \int_0^{\varphi} (1 - \sin^2\alpha \sin^2\theta)^{1/2} d\theta \tag{18.172}$$

or

$$E(x|m) = \int_0^{x} \left(\frac{1 - mt^2}{1 - t^2}\right)^{1/2} dt, \quad 0 \le m \le 1. \tag{18.173}$$

Again, for the case $\varphi = \pi/2,\ x = 1$, we have the **complete elliptic integral of the second kind**:

$$\begin{aligned} E(m) &= \int_0^{\pi/2} (1 - m\sin^2\theta)^{1/2} d\theta \\ &= \int_0^{1} \left(\frac{1 - mt^2}{1 - t^2}\right)^{1/2} dt, \quad 0 \le m \le 1. \end{aligned} \tag{18.174}$$

Series Expansions

For our range $0 \le m < 1$, the denominator of $K(m)$ may be expanded by the binomial series in Eq. (1.74):

$$(1 - m\sin^2\theta)^{-1/2} = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{(2n)!!} m^n \sin^{2n}\theta,$$

after which the resulting series is then integrated term by term. The integrals of the individual terms are beta functions (see Exercise 13.3.8), and we get

$$K(m) = \frac{\pi}{2}\left\{1 + \sum_{n=1}^{\infty}\left[\frac{(2n-1)!!}{(2n)!!}\right]^2 m^n\right\}. \tag{18.175}$$

Similarly (see Exercise 18.8.2),

$$E(m) = \frac{\pi}{2}\left\{1 - \sum_{n=1}^{\infty}\left[\frac{(2n-1)!!}{(2n)!!}\right]^2 \frac{m^n}{2n-1}\right\}. \tag{18.176}$$

These series can be identified as hypergeometric functions. Comparing with the general definitions in Section 18.5, we have

$$\begin{aligned} K(m) &= \frac{\pi}{2}\, {}_2F_1(\tfrac{1}{2}, \tfrac{1}{2}; 1; m), \\ E(m) &= \frac{\pi}{2}\, {}_2F_1(-\tfrac{1}{2}, \tfrac{1}{2}; 1; m). \end{aligned} \tag{18.177}$$

The complete elliptic integrals are plotted in Fig. 18.10.

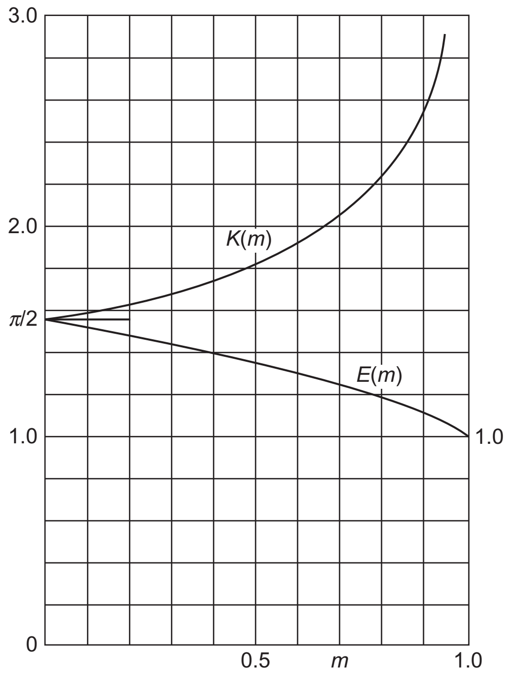

FIGURE 18.10 Complete elliptic integrals, $K(m)$ and $E(m)$.

Limiting Values

From the series Eqs. (18.175) and (18.176), or from the defining integrals,

$$\lim_{m\to 0} K(m) = \frac{\pi}{2}, \qquad \lim_{m\to 0} E(m) = \frac{\pi}{2}. \tag{18.178}$$

For $m \to 1$ the series expansions are of little use. However, the integrals yield

$$\lim_{m\to 1} K(m) = \infty, \qquad \lim_{m\to 1} E(m) = 1. \tag{18.179}$$

The divergence in $K(m)$ is logarithmic.

Elliptic integrals have been used extensively in the past for evaluating integrals. For instance, general integrals of the form

$$I = \int_0^x R\left(t, \sqrt{a_4t^4 + a_3t^3 + a_2t^2 + a_1t^1 + a_0}\right)dt,$$

where R is a rational function of its arguments, may be expressed in terms of elliptic integrals. Jahnke and Emde (Additional Readings) give pages of such transformations. With computers available for direct numerical evaluation, interest in these elliptic integral techniques has declined. A more extensive account of elliptic functions, integrals, and the related Jacobi theta functions can be found in Whittaker and Watson's treatise. Many formulas and tables of elliptic integrals are in AMS-55 and even more formulas are in Olver *et al*. (all of these sources are in the Additional Readings).

Exercises

18.8.1 The ellipse $x^2/a^2 + y^2/b^2 = 1$ may be represented parametrically by $x = a\sin\theta$, $y = b\cos\theta$. Show that the length of arc within the first quadrant is

$$a\int_0^{\pi/2}(1 - m\sin^2\theta)^{1/2}d\theta = aE(m).$$

Here $0 \le m = (a^2 - b^2)/a^2 \le 1$.

18.8.2 Derive the series expansion

$$E(m) = \frac{\pi}{2}\left\{1 - \left(\frac{1}{2}\right)^2\frac{m}{1} - \left(\frac{1\cdot 3}{2\cdot 4}\right)^2\frac{m^2}{3} - \cdots\right\}.$$

18.8.3 Show that

$$\lim_{m\to 0}\frac{(K - E)}{m} = \frac{\pi}{4}.$$

18.8.4 A circular loop of wire in the xy-plane, as shown in Fig. 18.11, carries a current I. Given that the vector potential is

$$A_\varphi(\rho,\varphi,z) = \frac{a\mu_0 I}{2\pi}\int_0^{\pi}\frac{\cos\alpha\, d\alpha}{(a^2 + \rho^2 + z^2 - 2a\rho\cos\alpha)^{1/2}},$$

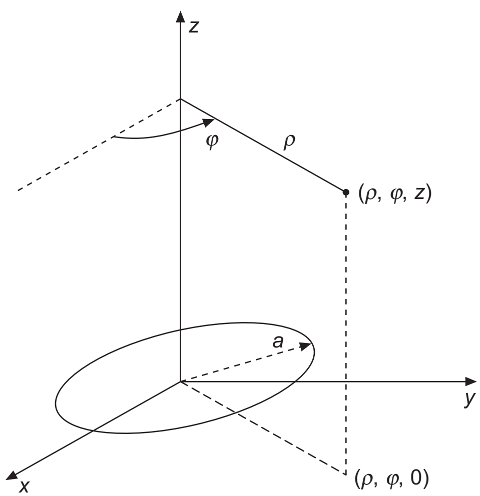

FIGURE 18.11 Circular wire loop.

show that

$$A_\varphi(\rho, \varphi, z) = \frac{\mu_0 I}{\pi k}\left(\frac{a}{\rho}\right)^{1/2}\left[\left(1 - \frac{k^2}{2}\right)K(k^2) - E(k^2)\right],$$

where

$$k^2 = \frac{4a\rho}{(a+\rho)^2 + z^2}.$$

Note. For extension of this exercise to **B**, see Smythe.[12]

18.8.5 An analysis of the magnetic vector potential of a circular current loop leads to the expression

$$f(k^2) = k^{-2}\left[(2 - k^2)K(k^2) - 2E(k^2)\right],$$

where $K(k^2)$ and $E(k^2)$ are the complete elliptic integrals of the first and second kinds. Show that for $k^2 \ll 1$ ($r \gg$ radius of loop)

$$f(k^2) \approx \frac{\pi k^2}{16}.$$

18.8.6 Show that

(a) $$\frac{dE(k^2)}{dk} = \frac{1}{k}(E - K),$$

(b) $$\frac{dK(k^2)}{dk} = \frac{E}{k(1-k^2)} - \frac{K}{k}.$$

Hint. For part (b) show that

$$E(k^2) = (1 - k^2)\int_0^{\pi/2}(1 - k\sin^2\theta)^{-3/2}d\theta$$

by comparing series expansions.

Additional Readings

Abramowitz, M., and I. A. Stegun, eds., *Handbook of Mathematical Functions*, Applied Mathematics Series-55 (AMS-55). Washington, DC: National Bureau of Standards (1964), paperback edition, Dover (1974). Chapter 22 is a detailed summary of the properties and representations of orthogonal polynomials. Other chapters summarize properties of Bessel, Legendre, hypergeometric, and confluent hypergeometric functions and much more. See also Olver *et al.*, below.

Buchholz, H., *The Confluent Hypergeometric Function*. New York: Springer Verlag (1953), translated (1969). Buchholz strongly emphasizes the Whittaker rather than the Kummer forms. Applications to a variety of other transcendental functions.

[12] W. R. Smythe, *Static and Dynamic Electricity*, 3rd ed. New York: McGraw-Hill (1969), p. 270.

Erdelyi, A., W. Magnus, F. Oberhettinger, and F. G. Tricomi, *Higher Transcendental Functions*, 3 vols. New York: McGraw-Hill (1953), reprinted, Krieger (1981). A detailed, almost exhaustive listing of the properties of the special functions of mathematical physics.

Fox, L., and I. B. Parker, *Chebyshev Polynomials in Numerical Analysis*. Oxford: Oxford University Press (1968). A detailed, thorough, but very readable account of Chebyshev polynomials and their applications in numerical analysis.

Gradshteyn, I. S., and I. M. Ryzhik, *Table of Integrals, Series and Products* (A. Jeffrey and D. Zwillinger, eds.), 7th ed. New York: Academic Press (2007).

Jahnke, E., and F. Emde, *Tables of Functions with Formulae and Curves*. Leipzig: Teubner (1933), Dover (1945).

Jahnke, E., F. Emde, and F. Lösch, *Tables of Higher Functions*, 6th ed. New York: McGraw-Hill (1960). An enlarged update of the work by Jahnke and Emde.

Lebedev, N. N., *Special Functions and Their Applications* (translated by R. A. Silverman). Englewood Cliffs, NJ: Prentice-Hall (1965), paperback, Dover (1972).

Luke, Y. L., *The Special Functions and Their Approximations*, 2 vols. New York: Academic Press (1969). Volume 1 is a thorough theoretical treatment of gamma functions, hypergeometric functions, confluent hypergeometric functions, and related functions. Volume 2 develops approximations and other techniques for numerical work.

Luke, Y. L., *Mathematical Functions and Their Approximations*. New York: Academic Press (1975). This is an updated supplement to *Handbook of Mathematical Functions with Formulas, Graphs and Mathematical Tables* (AMS-55).

Magnus, W., F. Oberhettinger, and R. P. Soni, *Formulas and Theorems for the Special Functions of Mathematical Physics*. New York: Springer (1966). An excellent summary of just what the title says.

Olver, F. W. J., D. W. Lozier, R. F. Boisvert, and C. W. Clark, eds., *NIST Handbook of Mathematical Functions*. Cambridge: Cambridge University Press (2010). Update of AMS-55 (Abramowitz and Stegun, above), but links to computer programs are provided instead of tables of data.

Rainville, E. D., *Special Functions*. New York: Macmillan (1960), reprinted, Chelsea (1971). This book is a coherent, comprehensive account of almost all the special functions of mathematical physics that the reader is likely to encounter.

Sansone, G., *Orthogonal Functions* (translated by A. H. Diamond). New York: Interscience (1959), reprinted, Dover (1991).

Slater, L. J., *Confluent Hypergeometric Functions*. Cambridge: Cambridge University Press (1960). This is a clear and detailed development of the properties of the confluent hypergeometric functions and of relations of the confluent hypergeometric equation to other ODEs of mathematical physics.

Sneddon, I. N., *Special Functions of Mathematical Physics and Chemistry*, 3rd ed. New York: Longman (1980).

Whittaker, E. T., and G. N. Watson, *A Course of Modern Analysis*. Cambridge: Cambridge University Press, reprinted (1997). The classic text on special functions and real and complex analysis.